Structural Analysis of Generalized Quasi-Cyclic LDPC Codes: Rank, Design, and Generator Matrices

Roxana Smarandache, *Senior Member, IEEE*, David G. M. Mitchell, *Senior Member, IEEE*, and Anthony Gómez-Fonseca, *Member, IEEE*

***Abstract*—Generalized low-density parity-check (GLDPC) codes, where single parity-check constraints on the code bits are replaced with generalized constraints (an arbitrary linear code), are a promising class of codes for low-latency communication. The block error rate performance of the GLDPC codes, combined with a complementary outer code, has been shown to outperform a variety of state-of-the-art code and decoder designs with suitable lengths and rates for the 5G ultra-reliable low-latency communication (URLLC) regime. A major drawback of these codes is that it is not known how to construct appropriate polynomial matrices to encode them efficiently. In this paper, we analyze practical constructions of quasi-cyclic GLDPC (QC-GLDPC) codes and show how to construct polynomial generator matrices in various forms using minors of the polynomial matrix. The approach can be applied to fully generalized matrices or partially generalized (with mixed constraint node types) to find better performance/rate trade-offs. The resulting encoding matrices are presented in useful forms that facilitate efficient implementation. The rich substructure displayed also provides us with new methods of determining low weight codewords, providing lower and upper bounds on the minimum distance and often giving those of weight equal to the minimum distance. Based on the minors of the polynomial parity-check matrix, we also give a formula for the rank of any parity-check matrix representing a QC-LDPC or QC-GLDPC code, and hence, the dimension of the code. Finally, we show that by applying double graph-liftings, the code parameters can be improved without affecting the ability to obtain a polynomial generator matrix.**

Index Terms—**LDPC codes, generalized LDPC codes, rank, minimum distance, generator matrices.**

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Roxana Smarandache is with the Department of Mathematics and the Department of Electrical Engineering, University of Notre Dame, Notre Dame, IN 46556 USA (e-mail: rsmarand@nd.edu).

David G. M. Mitchell is with the Klipsch School of Electrical and Computer Engineering, New Mexico State University, Las Cruces, NM 88003 USA (e-mail: dgmm@nmsu.edu).

Anthony Gómez-Fonseca was with the Department of Mathematics, University of Notre Dame, Notre Dame, IN 46556 USA. He is now with the Department of Mathematics and Statistics, University of South Florida, Tampa, FL 33620 USA (e-mail: agomezfonseca@usf.edu).

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I. Introduction

GENERALIZED low-density parity-check (GLDPC) codes, where single parity-check constraints are replaced by a general linear code, have been shown to have several advantages over conventional LDPC codes, including large minimum distance [1], [2], [3], good iterative decoding performance [4], [5], fast decoding convergence speed [6], and low error floors [7], [8]. The performance and convergence speed of the BP decoder can be further improved by applying a reinforcement learning approach to optimize the scheduling of messages in the decoder [9]. This improved performance from the generalized constraints comes at the cost of reduced coding rate, which is generally not desirable for many communication systems. However, these codes are well suited for next generation machine-to-machine (M2M) type communications and ultra reliable low latency communications (URLLC), which are expected to have low coding rate, e.g., $R = 1/12$, and short block lengths ranging from hundreds of bits to one or two thousand [5], [10], [11].

Conventional quasi-cyclic LDPC (QC-LDPC) codes have a highly structured parity-check matrix, which can be composed as an array of circulant matrices. Consequently, they are attractive for implementation purposes since their structure leads to efficiencies in decoder design [12]. Quasi-cyclic GLDPC (QC-GLDPC) codes were proposed in [7], where irregular protograph-based designs were shown to possess good performance in both the waterfall and error floor regions. In [5], a practical construction of QC-GLDPC codes were proposed for URLLC, where the optimal proportion of generalized Hamming constraints (to minimize the gap to capacity) was determined by an asymptotic analysis. The optimal proportions were found to be 75% of the constraint nodes in the $(2, 6)$-regular and $(2, 7)$-regular cases, while this proportion increases to 80% in the $(2, 15)$-regular case. A major drawback of these codes is that it is not known how to leverage the circulant based generalized matrix to construct appropriate polynomial matrices to encode them efficiently.

While conventional techniques to obtain generator matrices from LDPC matrices, such as Gauss-Jordan elimination and variants [13], can be applied to obtain generator matrices for QC-LDPC codes, these will typicaly result in dense and unstructured matrices. Instead, one can take advantage of the rich structure of QC-LDPC codes to obtain generator matrices. These include methods to construct a generator in standard form (e.g., [14]), which allows high throughput systematic encoding but the resulting generator matrix is typically dense.[1] Various other methods have been proposed that allow for sparse and structured generator matrices for QC-LDPC codes where there are specific constraints placed on the parity-check matrix [15], [16]. In a more recent paper, a general technique was proposed to form the generator matrix of a QC-LDPC code by computing the permanents of some sub-matrices of the parity-check matrix with appropriate modifications based on rank deficiencies [17].

The sparse parity-check matrix is often represented as an array of circulant permutations, which facilitate efficient implementation but will typically have a number of linearly dependent rows. Although Gaussian elimination can be employed to compute the rank with a complexity of $\mathcal{O}(n^3)$ in the number of variables n, it is desirable both from the perspective of computational complexity and from the perspective of yielding insight about the code structure to have an analytic way to compute the rank, particularly for classes of algebraic QC-GLDPC codes. Methods to compute the rank of QC-LDPC codes have been investigated, including approaches involving Fourier transforms [18] and the matrix polynomial representation [19]. However, these approaches are limited to certain code parameters. In [20], we presented a method to compute the rank of any parity-check matrix representing a QC-LDPC code, and hence the dimension of the code, by using the minors of the corresponding polynomial parity-check matrix.

In this paper, we first summarize and extend our results for QC-LDPC codes from [20], then show how the approach extends naturally to QC-GLDPC codes. Using the QC-GLDPC constructions from [5] and [7] as examples, and extending our work in [21], we demonstrate how to compute polynomial generator matrices, allowing efficient encoding. It is shown that the method also applies to mixed constructions, where the parity-check matrix has a mixture of generalized and single parity-check constraints. We consider the design of these codes and show that they can have large minimum distance. In particular, the generator matrices will often have rows that are equal (or close to) the minimum distance of the code, giving tight upper bounds that are hard to obtain otherwise. We discuss how the approach can also be used to determine the rank of the parity-check matrix of QC-GLDPC codes. Throughout the paper, we show that by applying double graph-liftings, the code parameters can be improved without affecting the ability to obtain a polynomial generator matrix. Importantly, this allows the code designer to finely tune the number of generalized constraints to optimize the performance/rate trade-off. Computer simulation results confirm the good performance promised by the constructed QC-GLDPC codes with high girth and large minimum distance.

[1] Systematic encoding, where the information is directly embedded in the codeword, is often preferred in practice.

The paper is organized as follows. Section II first provides some necessary background regarding QC-LDPC and QC-GLDPC codes. Sections III and IV establish theory regarding the rank and construction of polynomial generator matrices for QC-LDPC codes. Following this, we extend the results to QC-GLDPC codes in Section V. The paper is concluded in Section VI.

II. Definitions, Notation, and Background

An LDPC code $\mathcal{C}$ is described as the nullspace of a parity-check matrix H to which we associate a Tanner graph [22] in the usual way. The girth, denoted girth(H), of a graph is the length of the shortest cycle in the graph.

A. Protographs and Polynomial Representations

A protograph [23], [24] is a small bipartite graph represented by an $n_c \times n_v$ parity-check or *base* biadjacency matrix B with non-negative integer entries b_{ij}. The parity-check matrix H of a protograph-based LDPC block code can be created by replacing each non-zero entry b_{ij} by a sum of b_{ij} non-overlapping $N \times N$ permutation matrices and a zero entry by the $N \times N$ all-zero matrix. Graphically, this operation is equivalent to taking an N-fold graph cover, or "lifting", of the protograph. We denote the $N \times N$ circulant permutation matrix where the entries of the $N \times N$ identity matrix I are shifted to the left by r positions modulo N as I_r.

A QC-LDPC code of length $n = n_v N$ is a protograph-based LDPC code, for which the $N \times N$ lifting permutation matrices are all circulant matrices I_r. Thus a QC code has an $n_c N \times n_v N$ parity-check matrix H of the form

$$H = \begin{bmatrix} H_{1,1} & H_{1,2} & \cdots & H_{1,n_v} \\ \vdots & \vdots & \ddots & \vdots \\ H_{n_c,1} & H_{n_c,2} & \cdots & H_{n_c,n_v} \end{bmatrix}, \qquad (1)$$

where the $N \times N$ sub-matrices $H_{i,j}$ are circulant; applying equal circular shifts to each length N sub-block of a codeword results in a codeword. With the help of the well-known isomorphism between the ring of circulant matrices over the binary field $\mathbb{F}_2$ and the ring $\mathbb{F}_2[x]/(x^N - 1)$ of $\mathbb{F}_2$-polynomials modulo $x^N - 1$ (see, *e.g.*, [25]), a QC-LDPC code can also be described by an $n_c \times n_v$ polynomial parity-check matrix over $\mathbb{F}_2[x]/(x^N - 1)$. This polynomial description of QC-LDPC codes makes them attractive in practice, because they can be encoded efficiently using approaches like in [14] and decoded efficiently using belief-propagation-based decoding algorithms.

In particular, with the $n_c N \times n_v N$ parity-check matrix H described above we associate the polynomial parity-check matrix $H(x) \in \mathbb{F}_2^{\langle N \rangle}[x]^{n_c \times n_v}$ as

$$H(x) = \begin{bmatrix} h_{1,1}(x) & h_{1,2}(x) & \cdots & h_{1,n_v}(x) \\ \vdots & \vdots & \ddots & \vdots \\ h_{n_c,1}(x) & h_{n_c,2}(x) & \cdots & h_{n_c,n_v}(x) \end{bmatrix}. \qquad (2)$$

Here $h_{i,j}(x) = x^r$ corresponds to the circulant permutation matrix I_r. Moreover, with any vector

$$\mathbf{c} = (c_{1,0}, \ldots, c_{1,N-1}, \ldots, c_{n_v,0}, \ldots, c_{n_v,N-1})$$

in $\mathbb{F}_2^{n_v N}$, we associate the polynomial vector $\mathbf{c}(x) = \big(c_1(x), \ldots, c_{n_v}(x)\big)$ where $c_i(x) \triangleq \sum_{s=0}^{N-1} c_{i,s} x^s$. Then, the condition $H \cdot \mathbf{c}^\mathsf{T} = \mathbf{0}^\mathsf{T}$ (in $\mathbb{F}_2$) is equivalent to $H(x) \cdot \mathbf{c}(x)^\mathsf{T} = \mathbf{0}^\mathsf{T}$ in $\mathbb{F}_2[x]/(x^N - 1)$. For the remainder of this paper, H will be an $n_c N \times n_v N$ matrix over $\mathbb{F}_2$ that is an $n_c \times n_v$ array of $N \times N$ circulants and $H(x)$ will be the corresponding $n_c \times n_v$ polynomial matrix.

We use the following notation. For any positive integer L, $[L]$ denotes the set $\{1, 2, \ldots, L\}$. For any $n_c \times n_v$ matrix M, we let $M_{\mathcal{I},\mathcal{J}}$ be the sub-matrix of M that contains only the rows of M whose index appears in the set $\mathcal{I}$ and only the columns of M whose index appears in the set $\mathcal{J}$; if $\mathcal{I}$ equals the set of all row indices of M, *i.e.*, $\mathcal{I} = [n_c]$, we will simply write $M_{\mathcal{J}}$. We use the short-hand $M_{\mathcal{J}\setminus i}$ for $M_{\mathcal{J}\setminus\{i\}}$. If $\mathcal{I}$ and $\mathcal{J}$ have the same cardinality, we use $\Delta_{\mathcal{I},\mathcal{J}} = \det(M_{\mathcal{I},\mathcal{J}})$, and $\Delta_{\mathcal{J}} = \det(M_{[n_c],\mathcal{J}})$. To ensure clarity with the notation, we note that we will always work with either the block-circulant form of (1) or (more often) the polynomial representation (2). Consequently, we may interchangeably use the subscript set notation $H_{\mathcal{I},\mathcal{J}}$ or $H_{\mathcal{I},\mathcal{J}}(x)$, where $\mathcal{I} \subseteq [n_c]$ and $\mathcal{J} \subseteq [n_v]$. In the case $H_{\mathcal{I},\mathcal{J}}$, this would refer to an $N|\mathcal{I}| \times N|\mathcal{J}|$ submatrix of H corresponding to the associated $N \times N$ submatrices $H_{i,j}$, $i \in \mathcal{I}$, $j \in \mathcal{J}$; whereas $H_{\mathcal{I},\mathcal{J}}(x)$ would refer to the (corresponding) $|\mathcal{I}| \times |\mathcal{J}|$ submatrix of $H(x)$.

B. GLDPC Codes

Let a *component code* corresponding to one of the Nn_c constraints, c_i, be represented by a $p_i \times q_i$ parity-check matrix (where a conventional single parity-check (SPC) constraint corresponds to the $1 \times q_i$ all-ones matrix). The constraint matrix of a QC-GLDPC code is also represented as (1) and (2), however the full parity-check matrix is obtained by replacing each one entry with the corresponding column of the component parity-check matrix and zero entry with an all-zero column vector of the same dimension. Consequently, the design rate of the GLDPC code is

$$R = 1 - \frac{\sum_{i=1}^{n_c} p_i}{n_v}. \tag{3}$$

Conventionally, BP decoding is performed on the constraint graph, corresponding to (1), where component decoding can be performed as desired, e.g., BCJR, BP, etc., with different performance/complexity trade-offs. This paper focuses on the design and efficient encoding of GLDPC codes.

III. The Rank of QC Matrices

This section gives a simple formula for the rank of H and hence, the dimension of its associated code, that is a function of the full minors of the polynomial matrix $H(x)$. We note the paper [19] in which the authors had some particular cases based on the attempt to triangularize the matrix H. They do this only for the case $n_c = 3$. Our formula holds for any general matrix of size $n_c N \times n_v N$. In [20], we also gave a general triangular form into which any QC matrix can be written using elementary row operations and column permutations. Since these operations leave the dimension and the weight enumerator of the code unchanged, we can obtain an alternative way to compute the dimension of the code and thus completely solve the task of the paper [19].

The following theorem gives the formula for the rank over $\mathbb{F}_2$ of a scalar matrix using the computation of the degree of the minors of the corresponding polynomial matrix.

Theorem 1: We define $\gamma_0 \triangleq 1$, and, for all $i \in [n_c]$, we define the following in $\mathbb{F}_2[x]$:

$$\gamma_i(x) \triangleq \gcd\{\Delta_{\mathcal{I},\mathcal{J}}(H(x)) \mid |\mathcal{I}| = |\mathcal{J}| = i\},$$
$$d_i(x) \triangleq \gcd(\gamma_i(x)/\gamma_{i-1}(x), x^N + 1),$$

where, by assumption, $\gcd(0/0, x^N + 1) \triangleq \gcd(0, x^N + 1) = x^N + 1$. Then,

$$\operatorname{rank}_{\mathbb{F}_2}(H) = n_c \cdot N - \sum_{i=1}^{n_c} \deg d_i(x)$$
$$\Rightarrow \dim(\mathcal{C}) = (n_v - n_c) \cdot N + \sum_{i=1}^{n_c} \deg d_i(x).$$

Proof: In $\mathbb{F}_2[x]$, $H(x)$ is equivalent, after elementary row and column operations that leave its rank (but not its minimum distance) invariant, to its Smith normal form

$$\Gamma(x) \triangleq \left[\ \operatorname{diag}\left(\tfrac{\gamma_1}{\gamma_0}(x), \tfrac{\gamma_2}{\gamma_1}(x), \ldots, \tfrac{\gamma_{n_c}}{\gamma_{n_c-1}}(x)\right) \ \middle|\ 0_{n_c \times (n_v - n_c)}\ \right].$$

Equivalently, there exist invertible matrices $U(x), V(x)$ such that $U(x)H(x)V(x) = \Gamma(x)$ in $\mathbb{F}_2[x]$ and, therefore, such that $U(x)H(x)V(x) = \Gamma(x) \mod x^N - 1$. The rank of the binary $n_c N \times n_v N$ matrix H is thus equal to that of the binary $n_c N \times n_v N$ matrix Γ corresponding to the $n_c \times n_v$ polynomial matrix $\Gamma(x)$, which is equal to the sum of the ranks of the $N \times N$ circulant matrices corresponding to the polynomials $\gamma_i(x)/\gamma_{i-1}(x)$. Since each individual rank is $N - \deg(\gcd(\gamma_i(x)/\gamma_{i-1}(x), x^N + 1)) \triangleq N - \deg d_i(x)$, we obtain that

$$\operatorname{rank}_{\mathbb{F}_2}(H) = \operatorname{rank}_{\mathbb{F}_2}(\Gamma) = \sum_{i=1}^{n_c} (N - \deg d_i(x))$$
$$= n_c \cdot N - \sum_{i=1}^{n_c} \deg d_i(x)$$

and, equivalently, $\dim(\mathcal{C}) = (n_v - n_c) \cdot N + \sum_{i=1}^{n_c} \deg d_i(x)$. ■

Example 1: Let

$$H(x) \triangleq \begin{bmatrix} 1+x^2 & 1+x^4 & 1+x^6 & 1+x^8 & 1+x^{16} \\ x^4+x^{12} & x^{20}+x^{22} & x^{30}+x^{42} & x^{40}+x^{14} & 1+x^{50} \\ 1+x^4 & x^{30}+x^{24} & x^{12}+x^{14} & x^3+x^{13} & x+x^9 \end{bmatrix}$$

be a 3×5 polynomial matrix in $\mathbb{F}_2[x]$. Depending on N, its Tanner graph can have girth 6. We will keep N variable, and compute the rank of the $3N \times 5N$ matrix H obtained from $H(x)$ in the ring $\mathbb{F}_2[x]/(x^N + 1)$. Using the Magma program and its command for computing the needed determinants, $\gamma_i \triangleq \mathrm{GCD}(\mathrm{Minors}(H, i))$, in $\mathbb{F}_2[x]$, we obtain:

$$\gamma_1 = x^2 + 1, \quad \gamma_2 = x^4 + 1, \quad \gamma_3 = x^6 + x^4 + x^2 + 1,$$
$$\frac{\gamma_1}{\gamma_0} = \frac{\gamma_2}{\gamma_1} = \frac{\gamma_3}{\gamma_2} = x^2 + 1,$$

from which, for all $i \in [3]$, $d_i = \gcd(x^2+1, x^N+1) = \begin{cases} x^2+1, & \text{if } N \text{ even,} \\ x+1, & \text{if } N \text{ odd,} \end{cases}$ and so

$$d_i = \gcd(x^2+1, x^N+1) = \begin{cases} x^2+1, & \text{if } N \text{ even,} \\ x+1, & \text{if } N \text{ odd,} \end{cases} \Longrightarrow$$

the Smith Normal form of H, over $\mathbb{F}_2[x]$, is

$$\begin{bmatrix} x^2+1 & 0 & 0 & 0 & 0 \\ 0 & x^2+1 & 0 & 0 & 0 \\ 0 & 0 & x^2+1 & 0 & 0 \end{bmatrix}.$$

Therefore, the formula for the rank of H depends on the parity of N:

$$\operatorname{rank}(H) = 3(N - \deg(d_i)) = \begin{cases} 3N-6, & \text{if } N \text{ even,} \\ 3N-3, & \text{if } N \text{ odd.} \end{cases}$$

We observe that the same matrix gives a code of slightly larger dimension if N is even. For example, $N = 45$, gives H of rank 132 and the code of dimension 93, while $N = 46$ gives a matrix of the same rank, so a slightly longer code with dimension 98, while $N = 44$ gives H of rank 126 and a slightly shorter code of dimension 94. □

To conclude this section, we note that the complexity of Theorem 1 compares favorably to computing the rank via Gaussian elimination, which has a complexity of $\mathcal{O}(n^3)$ in the number of variables n. If $H \in \mathbb{F}_2[x]^{n_c \times n_v}$, the Smith normal form can be computed in $\mathcal{O}(n_c n_v r^{\omega-2} \log r)$ (see [26]), where r is the rank of H. This gives us the polynomials $\gamma_i(x)/\gamma_{i-1}(x)$ in Theorem 1. Now, we need to compute $d_i(x)$, which requires to compute the greatest common divisor between that polynomial and $x^N - 1$. Since these are polynomials of degree at most N, this takes $\mathcal{O}(N^2)$ operations in $\mathbb{F}_2$ using classical methods, or $\mathcal{O}(N \log^2 N \log(\log N))$ operations in $\mathbb{F}_2$ using fast methods [27].

IV. Generator Matrices for QC-LDPC Codes

In [20] we gave a method (based on polynomial minors of $H(x)$) to construct polynomial generator matrices for QC-LDPC codes. We extend these results and show how to obtain sets of codewords that can be used to create generator matrices for QC codes. We extend here the simple technique described in [28] to construct codewords of codes described by polynomial parity-check matrices.[2] The following two lemmas allow us to construct codewords and generator matrices, and thus, to give upper bound on the minimum distance of the code.

Lemma 1: Let $\mathcal{S} = \{i_1, i_2, \ldots, i_{n_c+1}\}$ be an arbitrary size (n_c+1) subset of $[n_v]$. Let $\mathbf{c}_{\mathcal{S}}(x)$ be the vector defined as

$$\mathbf{c}_{\mathcal{S}}(x) = \left(c_{\mathcal{S},1}(x), \ldots, c_{\mathcal{S},n_v}(x)\right) =$$
$$\Big(0\cdots0 \ \underset{\substack{\uparrow \\ i_1}}{\Delta^{\mathsf{T}}_{\mathcal{S}\setminus i_1}} \ 0\cdots0 \ \underset{\substack{\uparrow \\ i_2}}{\Delta^{\mathsf{T}}_{\mathcal{S}\setminus i_2}} \ 0\cdots0 \ \cdots \ \underset{\substack{\uparrow \\ i_{n_c+1}}}{\Delta^{\mathsf{T}}_{\mathcal{S}\setminus i_{n_c+1}}} \ 0\cdots0\Big)$$

i.e., $c_{\mathcal{S},i}(x) \triangleq \det^{\mathsf{T}}\left(H_{\mathcal{S}\setminus i}(x)\right) = \Delta_{\mathcal{S}\setminus i}$ if $i \in \mathcal{S}$ and $c_{\mathcal{S},i}(x) \triangleq 0$ otherwise. Then $\mathbf{c}_{\mathcal{S}}(x)$ and $\frac{1}{a(x)}\mathbf{c}_{\mathcal{S}}(x)$ are codewords in $\mathcal{C}$, for every $a(x)$ a divisor of $\Delta_{\mathcal{S}\setminus i}$ in $\mathbb{F}_2[x]$, for all $i \in \mathcal{S}$, where the vectors (and the divisions) are first computed over $\mathbb{F}_2[x]$ and then projected onto $\mathbb{F}_2[x]/(x^N+1)$.

The vectors $\mathbf{c}_{\mathcal{S}}(D)$ and $\frac{1}{a(D)}\mathbf{c}_{\mathcal{S}}(D)$ are also codewords in the time-invariant convolutional code generated by $H(D) = H(x)\big|_{x=D}$.

Proof: Let $\mathbf{s}^{\mathsf{T}}(x) = H(x) \cdot \mathbf{c}^{\mathsf{T}}_{\mathcal{S}}(x)$ be the syndrome. Then, for any $j \in [n_c]$,

$$s_j(x) = \sum_{i \in \mathcal{S}} h_{j,i}(x) \cdot \det\left(H_{\mathcal{S}\setminus i}(x)\right)$$
$$= \begin{vmatrix} h_{j,i_1}(x) & h_{j,i_2}(x) & \cdots & h_{j,i_{n_c}+1}(x) \\ \hline h_{1,i_1}(x) & h_{1,i_2}(x) & \cdots & h_{1,i_{n_c}+1}(x) \\ \vdots & \vdots & & \vdots \\ h_{n_c,i_1}(x) & h_{n_c,i_2}(x) & \cdots & h_{n_c,i_{n_c}+1}(x) \end{vmatrix} = 0,$$

since $s_j(x)$ is the co-factor expansion of the determinant of a $|\mathcal{S}| \times |\mathcal{S}|$ matrix which has determinant zero over $\mathbb{F}_2[x]$.

Since $a(x)$ is a divisor of all of the components, we can divide the codeword $\mathbf{c}_{\mathcal{S}}(x)$ by $a(x)$. Its syndrome will be the syndrome of $\mathbf{c}_{\mathcal{S}}(x)$ divided by $a(x)$, which is still zero in $\mathbb{F}_2[x]$. Therefore, $\mathbf{c}_{\mathcal{S}}(x)$ and $\frac{1}{a(x)}\mathbf{c}_{\mathcal{S}}(x)$ are codewords in the associated convolutional code, and therefore, when projected onto $\mathbb{F}_2[x]/(x^N+1)$, they are also codewords in the QC code.

Finally, since $H(x) \cdot \mathbf{c}^{\mathsf{T}}_{\mathcal{S}}(x) = 0$ in $\mathbb{F}_2[x]$ and the division by $a(x)$ is performed in $\mathbb{F}_2[x]$, we obtain that $H(D) \cdot \mathbf{c}^{\mathsf{T}}_{\mathcal{S}}(D) = 0$ and, therefore, the vectors $\mathbf{c}_{\mathcal{S}}(D)$ and $\frac{1}{a(D)}\mathbf{c}_{\mathcal{S}}(D)$ are also codewords in the time-invariant convolutional code generated by $H(D) = H(x)\big|_{x=D}$. ■

Lemma 2: Let $s \in \{0, \ldots, n_c-1\}$, and let $\mathcal{S} = \{i_1, i_2, \ldots, i_{s+1}\}$ be an arbitrary size $(s+1)$ subset of $[n_v]$. Let $\mathcal{T} = \{j_1, j_2, \ldots, j_s\}$ be an arbitrary size s subset of $[n_c]$ (where a size 0 set is an empty set), and let

$$\mathbf{c}_{\mathcal{T},\mathcal{S}} = \left(c_{\mathcal{T},\mathcal{S},1}(x), \ldots, c_{\mathcal{T},\mathcal{S},n_v}(x)\right) =$$
$$\Big(0\cdots0 \ \underset{\substack{\uparrow \\ i_1}}{(f(x)\Delta_{\mathcal{T},\mathcal{S}\setminus i_1})^{\mathsf{T}}} \ \cdots \ \underset{\substack{\uparrow \\ i_{s+1}}}{(f(x)\Delta_{\mathcal{T},\mathcal{S}\setminus i_{s+1}})^{\mathsf{T}}} \ 0\cdots0\Big),$$

i.e., $c_{\mathcal{T},\mathcal{S},i}(x) \triangleq (f(x)\Delta_{\mathcal{T},\mathcal{S}\setminus i})^{\mathsf{T}}$ if $i \in \mathcal{S}$ and $c_{\mathcal{T},\mathcal{S},i}(x) \triangleq 0$ otherwise. Then $\mathbf{c}_{\mathcal{T},\mathcal{S}}(x)$ is a codeword in $\mathcal{C}$ if and only if $f(x) \cdot \Delta_{\mathcal{T}\cup j,\mathcal{S}} = 0$, for all $j \in [n_c] \setminus T$.

Proof: Let $\mathbf{s}^{\mathsf{T}}(x) = H(x) \cdot \mathbf{c}^{\mathsf{T}}_{\mathcal{T},\mathcal{S}}(x)$ be the syndrome. Then, for any $j \in [n_c]$,

$$s_j(x) = \sum_{i \in \mathcal{S}} h_{j,i}(x) \cdot f(x) \cdot \det\left(H_{\mathcal{T},\mathcal{S}\setminus i}(x)\right)$$
$$= f(x) \begin{vmatrix} h_{j,i_1}(x) & h_{j,i_2}(x) & \cdots & h_{j,i_s+1}(x) \\ \hline h_{j_1,i_1}(x) & h_{j_1,i_2}(x) & \cdots & h_{j_1,i_{s+1}}(x) \\ \vdots & \vdots & & \vdots \\ h_{j_s,i_1}(x) & h_{j_s,i_2}(x) & \cdots & h_{j_s,i_{s+1}}(x) \end{vmatrix}$$
$$= f(x) \cdot \begin{cases} 0, & \text{if } j \in \mathcal{T}, \\ \Delta_{\mathcal{T}\cup j,\mathcal{S}}, & \text{if } j \in [n_c] \setminus \mathcal{T} \end{cases}$$

[2]This is, in turn, an extension of a codeword construction technique by MacKay and Davey [29, Theorem 2].

$$= \begin{cases} 0, & \text{if } j \in \mathcal{T}, \\ f(x) \cdot \Delta_{\mathcal{T} \cup j, \mathcal{S}}, & \text{if } j \in [n_c] \setminus \mathcal{T}. \end{cases}$$

Therefore, we obtain that $\mathbf{s}(x) = \mathbf{0}$ and, thus, $\mathbf{c}_{\mathcal{T},\mathcal{S}}(x)$ is a codeword if and only if $f(x) \cdot \Delta_{\mathcal{T} \cup j, \mathcal{S}} = 0$ for all $j \in [n_c] \setminus T$. ■

We will use these two lemmas to construct codewords and generator matrices for QC codes. We will give an example first.

Example 2: Let

$$H_1(x) = \begin{bmatrix} 1+x & 1+x^2 & (1+x^3)(1+x) & 1+x^3 \end{bmatrix}.$$

We apply Lemma 1 and obtain the following 12 codewords with $a(x) = x + 1$:

$$\begin{aligned}
\mathbf{c}_{\{1,2\}} &= (1+x^2, 1+x, 0, 0), \\
\frac{1}{x+1}\mathbf{c}_{\{1,2\}} &= (1+x, 1, 0, 0), \\
\mathbf{c}_{\{1,3\}} &= ((1+x^3)(1+x), 0, 1+x, 0), \\
\frac{1}{x+1}\mathbf{c}_{\{1,3\}} &= (1+x^3, 0, 1, 0), \\
\mathbf{c}_{\{1,4\}} &= (1+x^3, 0, 0, 1+x), \\
\frac{1}{x+1}\mathbf{c}_{\{1,4\}} &= (1+x+x^2, 0, 0, 1), \\
\mathbf{c}_{\{2,3\}} &= (0, (1+x^3)(1+x), 1+x^2, 0), \\
\frac{1}{x^2+1}\mathbf{c}_{\{2,3\}} &= (0, 1+x+x^2, 1, 0), \\
\mathbf{c}_{\{2,4\}} &= (0, 1+x^3, 0, 1+x^2), \\
\frac{1}{x+1}\mathbf{c}_{\{2,4\}} &= (0, 1+x+x^2, 0, 1+x), \\
\mathbf{c}_{\{3,4\}} &= (0, 0, 1+x^3, (1+x^3)(1+x)), \\
\frac{1}{x+1}\mathbf{c}_{\{3,4\}} &= (0, 0, 1, 1+x).
\end{aligned}$$

We apply Lemma 2, for $s = 0$, and obtain the following 4 codewords:

$$\begin{aligned}
&(f_1(x), 0, 0, 0), \text{where } f_1(x)(1+x) = 1+x^N, \text{in } \mathbb{F}_2[x], \\
&(0, f_2(x), 0, 0), \text{where } f_2(x) \cdot \gcd(1+x^2, 1+x^N) \\
&\qquad\qquad = 1+x^N, \text{in } \mathbb{F}_2[x], \\
&(0, 0, f_3(x), 0), \text{where } f_3(x) \cdot \gcd(1+x+x^3+x^4, 1+x^N) \\
&\qquad\qquad = 1+x^N, \text{in } \mathbb{F}_2[x], \\
&(0, 0, 0, f_4(x)), \text{where } f_4(x) \cdot \gcd(1+x^3, 1+x^N) \\
&\qquad\qquad = 1+x^N, \text{in } \mathbb{F}_2[x].
\end{aligned}$$

We note that $(0, f_1(x), 0, 0), (0, 0, f_1(x), 0), (0, 0, 0, f_1(x))$ are also codewords.

Since $\mathrm{rank}(H) = N - \deg(1+x) = N - 1$, the rank of G needs to be $3N + 1$. So in order to achieve the desired rank, we need to use some of the 12 codewords in the top list, together with one or more of the 4 codewords from the bottom list, depending on N. (This will be elaborated on below after we introduce Theorem 2.) For example,

$$G_1(x) \triangleq \left[\begin{array}{cccc}
1+x^2 & 1+x & 0 & 0 \\
(1+x^3)(1+x) & 0 & 1+x & 0 \\
1+x^3 & 0 & 0 & 1+x \\
\hline
f_1(x) & 0 & 0 & 0 \\
0 & f_2(x) & 0 & 0 \\
0 & 0 & f_3(x) & 0 \\
0 & 0 & 0 & f_4(x)
\end{array}\right],$$

$$G_2(x) \triangleq \left[\begin{array}{cccc}
1+x & 1 & 0 & 0 \\
1+x^3 & 0 & 1 & 0 \\
1+x+x^2 & 0 & 0 & 1 \\
\hline
f_1(x) & 0 & 0 & 0
\end{array}\right]$$

are two generator matrices for this code. In this case, the second matrix has lower weight rows, so it would seem to be the natural choice; however, it is not always the case that the second type will be more attractive, since when we divide a polynomial by another we do not always obtain a lower weight. In fact, sometimes, the weight increases drastically, for example when we divide $1 + x^N$ by $1 + x$. Therefore, both matrices could be considered and compared. They give different sets of codewords, and thus upper bounds on the minimum distance of the code. In this case, we obtain $d(\mathcal{C}) \leq 3$ from the second matrix. □

As we saw in Example 2, the two lemmas can be used to obtain a variety of generator matrices, with the goal of getting one of the lowest weight, equal to the minimum distance.

Theorem 2: Suppose there exists a subset $\mathcal{S}$ of size n_c of $[n_v]$ such that $\gcd(\Delta_{\mathcal{S}}, x^N + 1) = 1$, i.e., $\Delta_{\mathcal{S}}$ is invertible in $\mathbb{F}_2[x]/(x^N + 1)$. For simplicity, and w.l.o.g., we assume that $\mathcal{S} = [n_c]$. For all $i \in [m]$, $m = n_v - n_c$, let $S_i = S \cup \{n_c + i\}$, and let G be the $m \times n_v$ matrix defined as

$$G \triangleq \begin{bmatrix} \mathbf{c}_{S_1} \\ \mathbf{c}_{S_2} \\ \vdots \\ \mathbf{c}_{S_m} \end{bmatrix} = \begin{bmatrix}
\Delta^{\mathsf{T}}_{S_1 \setminus 1} & \cdots & \Delta^{\mathsf{T}}_{S_1 \setminus n_c} & \Delta^{\mathsf{T}}_{\mathcal{S}} & 0 & \cdots & 0 \\
\Delta^{\mathsf{T}}_{S_2 \setminus 1} & \cdots & \Delta^{\mathsf{T}}_{S_2 \setminus n_c} & 0 & \Delta^{\mathsf{T}}_{\mathcal{S}} & \cdots & 0 \\
\vdots & & \vdots & \vdots & \vdots & & \vdots \\
\Delta^{\mathsf{T}}_{S_m \setminus 1} & \cdots & \Delta^{\mathsf{T}}_{S_m \setminus n_c} & 0 & 0 & \cdots & \Delta^{\mathsf{T}}_{\mathcal{S}}
\end{bmatrix}.$$

Then, G and $(\Delta^{\mathsf{T}}_{\mathcal{S}})^{-1} G$ are generator matrices for $\mathcal{C}$, where the second one is in standard form.

Proof: For all $i \in [m]$, $m = n_v - n_c$, we apply Lemma 1 to each of the sets $S_i = S \cup \{n_c + i\}$, to obtain the linearly independent set $\{\mathbf{c}_{S_1}, \mathbf{c}_{S_2}, \cdots, \mathbf{c}_{S_m}\}$ of $n_v - n_c$ rows of G, of rank $mN = n_v N - n_c N = \dim(C)$, as needed. The second matrix has the same property. ■

For brevity in examples, we will often drop the set notation in subscripts where there is no ambiguity and single digit integers. For example, if $S = \{1, 2, 3\}$, we write $\Delta_{123} = \det(H_{123})$.

Example 3: (AR4JA codes.) Let $H(x)$ be the polynomial parity-check matrix given by

$$H(x) = \begin{bmatrix} 0 & 0 & 1 & 0 & 1+x \\ 1 & 1 & 0 & 1 & x+x^2+x^3 \\ 1 & x+x^2 & 0 & 1+x^3 & 1 \end{bmatrix},$$

with $N = 4$. The rows of the matrix $G(x)$ below are linearly independent codewords for the code $\mathcal{C}$ given by H. Since $\Delta_{123} = \det(H_{123}) = x^2 + x + 1$ is irreducible in $\mathbb{F}_2[x]/(x^4 + 1)$, the matrix $G(x)$ has rank $2N = 8$ and thus forms a generator matrix for the code $\mathcal{C}$, where

$$G(x) = \begin{bmatrix} \Delta_{234}^{\mathsf{T}} & \Delta_{134}^{\mathsf{T}} & \Delta_{124}^{\mathsf{T}} & \Delta_{123}^{\mathsf{T}} & 0 \\ \Delta_{235}^{\mathsf{T}} & \Delta_{135}^{\mathsf{T}} & \Delta_{125}^{\mathsf{T}} & 0 & \Delta_{123}^{\mathsf{T}} \end{bmatrix} =$$
$$\begin{bmatrix} (x+1)^3 & x & 0 & x^3+x^2+1 & 0 \\ x^3+x^2+1 & (x+1)^3 & x+1 & 0 & x^3+x^2+1 \end{bmatrix}.$$

The vectors

$$(\Delta_{245}^{\mathsf{T}}, \Delta_{145}^{\mathsf{T}}, 0, \Delta_{125}^{\mathsf{T}}, \Delta_{124}^{\mathsf{T}}),$$
$$(\Delta_{345}^{\mathsf{T}}, 0, \Delta_{145}^{\mathsf{T}}, \Delta_{135}^{\mathsf{T}}, \Delta_{134}^{\mathsf{T}}),$$
$$(0, \Delta_{345}^{\mathsf{T}}, \Delta_{245}^{\mathsf{T}}, \Delta_{235}^{\mathsf{T}}, \Delta_{234}^{\mathsf{T}}),$$

are also codewords, where $\Delta_{145} = \det(H_{145}) = x+1$, $\Delta_{245} = \det(H_{245}) = 0$, and $\Delta_{345} = \det(H_{345}) = x$, but they are linear combinations of the ones displayed in the matrix $G(x)$, so they are not needed for the generator matrix. Displaying them is useful, however, since the nonzero minimum of the weights of the five codewords gives an upper bound on the minimum distance of the code. In this case, the codeword $(\Delta_{245}^{\mathsf{T}}, \Delta_{145}^{\mathsf{T}}, 0, \Delta_{125}^{\mathsf{T}}, \Delta_{124}^{\mathsf{T}}) = (0, x+1, 0, x+1, 0)$ has weight 4, which is in fact the minimum distance of the code; this code has parameters $[20, 8, 4]$.

Multiplying the two rows of G by $(x^3+x^2+1)^{-1} = x^2+x+1$ gives a generator matrix in standard form,

$$G(x) = \begin{bmatrix} x^3+x^2+x+1 & x^3+x^2+x & 0 & 1 & 0 \\ 1 & x^3+x^2+x+1 & x^3+1 & 0 & 1 \end{bmatrix},$$

while

$$H(x) = \begin{bmatrix} 1 & 0 & 0 & x^3+x^2+x+1 & 1 \\ 0 & 1 & 0 & x+x^2+x^3 & x^3+x^2+x+1 \\ 0 & 0 & 1 & 0 & x+1 \end{bmatrix}$$

is a polynomial parity-check matrix in systematic form. □

Remark 1: In Example 2, we saw how to extend Theorem 2 using Lemma 2 to obtain a generator matrix in the case $\gcd(\gamma_{n_c}(x), x^N+1) = g(x) \neq 1$, for $\gamma_{n_c}(x) = \gcd\{\Delta_{\mathcal{I},\mathcal{J}}(H(x)) \mid |\mathcal{I}| = |\mathcal{J}| = n_c\}$. We choose $f(x)$ such that $f(x)g(x) = x^N+1$ in $\mathbb{F}_2[x]$, and apply Lemma 2 to, for example, the sets $[n_c] \setminus i, \mathcal{S}$, to obtain extra vectors, $\mathbf{c}_{[n_c]\setminus i,\mathcal{S}}(x)$, that can be added to the matrix G, of rank $m(N - \deg g(x))$, to increase the rank to the dimension of $\mathcal{C}$, which can be computed using Theorem 1 as $\dim(\mathcal{C}) = m \cdot N + \sum_{i=1}^{n_c} \deg d_i(x)$.

For example, if we divide each of the rows of G by $\gamma_{n_c}(x)$ in $\mathbb{F}_2[x]$ and then add the codewords (4), as shown at the bottom of the page, we obtain a generator matrix for the code. Without dividing, the top part of the matrix obtained from Theorem 2 has a lower rank than the one when we divide, so we would need to add more codewords obtained from Lemma 2, possibly all (as is the case in Example 2), to increase the rank of the matrix to equal the dimension. □

In Section II-B, we will show how to obtain generator matrices for QC-GLDPC codes and we will encounter this situation several times. In addition, we will need the theorem below, which simplifies our task, by allowing us to reduce finding a generator matrix for an $[n_v N, n_v N - \mathrm{rank}(H)]$ code to finding a generator matrix for a shorter code of the same dimension.

Theorem 3: Let $H_{short} \triangleq H_{m_1 \times n}$ be an $m_1 \times n$ matrix and $A \triangleq A_{m_2 \times n}$ be an $m_2 \times n$ matrix. Let

$$H \triangleq H_{(m_1+m_2)\times(n+m_2)} \triangleq \begin{bmatrix} H_{short} & 0_{m_1 \times m_2} \\ A & I_{m_2 \times m_2} \end{bmatrix},$$

G_{short} be a generator matrix for $\ker(H_{short})$, and $g_{GC} \triangleq \begin{bmatrix} I_{n\times n} & A^{\mathsf{T}} \end{bmatrix}$ is a generator matrix for the code $\ker \begin{bmatrix} A & I_{m_2 \times m_2} \end{bmatrix}$ in standard form. Then the matrix

$$G \triangleq \begin{bmatrix} G_{short} & G_{short}A^{\mathsf{T}} \end{bmatrix} = G_{short} \begin{bmatrix} I_{n\times n} & A^{\mathsf{T}} \end{bmatrix}$$
$$= G_{short} \cdot g_{GC}$$

is a generator matrix for $\ker(H)$, where the codewords $(\mathbf{v}_1, \mathbf{v}_2)$, satisfy

$$(\mathbf{v}_1, \mathbf{v}_2) = (\mathbf{v}_1, \mathbf{v}_1 A^{\mathsf{T}}) = \mathbf{v}_1 \begin{bmatrix} I_{n\times n} & A^{\mathsf{T}} \end{bmatrix} = \mathbf{v}_1 \cdot g_{GC},$$

and

$$d_{min}(\ker(H)) \geq d_{min}(\ker(H_{short})).$$

Proof: Let $(\mathbf{v}_1, \mathbf{v}_2)^{\mathsf{T}} \in \ker(H)$, therefore, $\mathbf{v}_1^{\mathsf{T}} \in \ker(H_{short})$, and $\mathbf{v}_2 = \mathbf{v}_1 A^{\mathsf{T}}$. We have that

$$HG^{\mathsf{T}} = \begin{bmatrix} H_{short} & \mathbf{0} \\ A & I \end{bmatrix} \begin{bmatrix} G_{short}^{\mathsf{T}} \\ (G_{short}A^{\mathsf{T}})^{\mathsf{T}} \end{bmatrix}$$
$$= \begin{bmatrix} H_{short}G_{short}^{\mathsf{T}} \\ AG_{short}^{\mathsf{T}} + (G_{short}A^{\mathsf{T}})^{\mathsf{T}} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

since $H_{short}G_{short}^{\mathsf{T}} = 0$. Further, since

$$\begin{aligned} &(n+m_2) - \mathrm{rank}(H) \\ &= (n+m_2) - (\mathrm{rank}(H_{short}) + \mathrm{rank}(I_{m_2\times m_2})) \\ &= (n+m_2) - (\mathrm{rank}(H_{short}) + m_2) = n - \mathrm{rank}(H_{short}) \\ &= \mathrm{rank}(G_{short}) = \mathrm{rank}(G), \end{aligned}$$

we obtain that G is a generator matrix for H.

Let $\mathbf{v}_1$ be a codeword of the smallest weight in $\ker(H_{short})$, then

$$\begin{aligned} d_{min}(\ker(H)) \geq wt(\mathbf{v}_1, \mathbf{v}_2) &\geq wt(\mathbf{v}_1) \\ &= d_{min}(\ker(H_{short})), \end{aligned}$$

and the lower bound on the minimum distance follows. ■

V. Generator Matrices for QC-GLDPC Codes

In this section, we construct generalized LDPC codes directly and also by using pre-lifting [30], based on results developed in [5] that are likely to have good performance, and show how we can find polynomial generator matrices for them.

$$\begin{bmatrix} (f(x)\Delta_{[n_c]\setminus 1,\mathcal{S}})^{\mathsf{T}} & \cdots & (f(x)\Delta_{[n_c]\setminus 1,\mathcal{S}})^{\mathsf{T}} & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & & \vdots \\ (f(x)\Delta_{[n_c]\setminus n_c,\mathcal{S}})^{\mathsf{T}} & \cdots & (f(x)\Delta_{[n_c]\setminus n_c,\mathcal{S}})^{\mathsf{T}}\mathrm{diag}_m(\Delta_{\mathcal{S}})^{\mathsf{T}} & 0 & \cdots & 0 \end{bmatrix}. \tag{4}$$

We start from the general class of $2N \times n_v N$ QC-GLDPC codes based on the all-one protograph, and give examples for $n_v = 6, 7, 15$. Therefore, we can take, w.l.o.g., the following constraint matrix

$$H(x) \triangleq \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & x^{i_2} & x^{i_3} & \cdots & x^{i_{n_v}} \end{bmatrix}. \quad (5)$$

We remark that conditions were given in [30] for necessary and sufficient conditions on the exponents $i_2, i_3, \ldots, i_{n_v}$ to achieve a desired girth. For example, the girth is 4 if and only if there are at least 2 exponents that are equal (modulo N).

A. Direct Generalization of One Constraint

We consider first the case where the first constraint (N check nodes) are all simple nodes (single parity-checks) and the second constraint node is a general code with a given parity-check matrix.

Corollary 1: Let $H(x)$ be a $2 \times n_v$ polynomial constraint matrix like in (5) and let $h_{GC} = \begin{bmatrix} M_{n\times m} & I_{n\times n} \end{bmatrix}$ be the parity-check matrix for the constraint code in systematic form, where M is an $n \times m$ matrix, I is the $n \times n$ identity matrix, and $n_v = m + n$. Let H_{GC} be the parity-check matrix of the overall GLDPC code based on H and h_{GC}, as

$$H_{GC} \triangleq \left[\begin{array}{cccc|ccc} 1 & x^{i_2} & \cdots & x^{i_m} & x^{i_{m+1}} & \cdots & x^{i_{n_v}} \\ \hline & & M & & & I & \end{array}\right]. \quad (6)$$

Let $H_{short} \triangleq \begin{bmatrix} f_1 & \cdots & f_m \end{bmatrix} \triangleq \begin{bmatrix} 1 & x^{i_2} & \cdots & x^{i_m} \end{bmatrix} + \begin{bmatrix} x^{i_{m+1}} & \cdots & x^{i_{n_v}} \end{bmatrix} M$ and let G_{short}, G'_{short} be the matrices

$$G_{short} = \left[\begin{array}{ccccc} f_m^{\mathsf{T}} & 0 & \cdots & 0 & f_1^{\mathsf{T}} \\ 0 & f_m^{\mathsf{T}} & \cdots & 0 & f_2^{\mathsf{T}} \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & f_m^{\mathsf{T}} & f_{m-1}^{\mathsf{T}} \\ \hline f^{\mathsf{T}} & 0 & \cdots & 0 & 0 \\ 0 & f^{\mathsf{T}} & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & f^{\mathsf{T}} \end{array}\right], \quad (7)$$

$$G'_{short} = \left[\begin{array}{ccccc} \left(\frac{f_m}{g(x)}\right)^{\mathsf{T}} & 0 & \cdots & 0 & \left(\frac{f_1}{g(x)}\right)^{\mathsf{T}} \\ 0 & \left(\frac{f_m}{g(x)}\right)^{\mathsf{T}} & \cdots & 0 & \left(\frac{f_2}{g(x)}\right)^{\mathsf{T}} \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & \left(\frac{f_m}{g(x)}\right)^{\mathsf{T}} & \left(\frac{f_{m-1}}{g(x)}\right)^{\mathsf{T}} \\ \hline 0 & 0 & \cdots & 0 & f^{\mathsf{T}} \end{array}\right], \quad (8)$$

where $g(x)$ is defined as $f(x)g(x) = x^N + 1$, with $\gcd(f_1, \ldots, f_m, x^N + 1) = g(x)$.[3]

Then, G_{short} and G'_{short} are generator matrices for $\ker(H_{short})$ and the matrices

$$G = \left[\begin{array}{c|c} G_{short} & G_{short}M^{\mathsf{T}} \end{array}\right], G' = \left[\begin{array}{c|c} G'_{short} & G'_{short}M^{\mathsf{T}} \end{array}\right]$$

are generator matrices for $\ker(H_{GC})$.

[3]We need to have $\gcd(f_m, x^N + 1) = g(x)$ in $\mathbb{F}_2[x]$ (and not a larger degree polynomial), otherwise, we choose another f_i that satisfies $\gcd(f_i, x^N + 1) = g(x)$ to be on the diagonal part of the matrix above. Also, if $g(x) = 1$, then $f(x) = 0$ and the last $m - 1$ rows are omitted.

Proof: After row operations on (6), we obtain a row-equivalent matrix, which is a parity-check matrix for the same code, as

$$H_{GC} \sim \left[\begin{array}{c|c} \begin{bmatrix} 1 & x^{i_2} & \cdots & x^{i_m} \end{bmatrix} + \begin{bmatrix} x^{i_{m+1}} & \cdots & x^{i_{n_v}} \end{bmatrix} M & \mathbf{0} \\ \hline M & I \end{array}\right]$$
$$\sim \left[\begin{array}{ccc|c} f_1 & \cdots & f_m & \mathbf{0} \\ \hline & M & & I \end{array}\right]$$

We now apply Theorem 3 for $H_{short} \triangleq \begin{bmatrix} f_1 & \cdots & f_m \end{bmatrix}$, then Theorem 2, Remark 1, and Lemma 2, to obtain the generator matrix G_{short} as in the statement.

Finally, we apply Theorem 1 to compute the ranks of G_{short} and H_{short} and show that they are correctly related, where

$$\begin{aligned} \operatorname{rank}(H_{short}) &= N - \deg f(x) = \deg g(x), \\ \operatorname{rank}(G_{short}) &= (m-1)(N - \deg f(x)) + m(N - \deg g(x)) \\ &= (m-1)(2N - \deg f(x) - \deg f(x)) \\ &\quad + (N - \deg g(x)) \\ &= mN - \deg g(x) = mN - \operatorname{rank}(H_{short}), \end{aligned}$$

and

$$\begin{aligned} \operatorname{rank}(G'_{short}) &= (m-1)N + N - \deg g(x)) \\ &= (m-1)N + \deg f(x) \\ &= mN - (N - \deg(f(x)) \\ &= mN - \operatorname{rank}(H_{short}). \end{aligned}$$

Therefore, G_{short} and G'_{short} are generator matrices for H_{short}. Following Theorem 3, the matrices G and G' form generator matrices for $\ker(H_{GC})$. ■

Remark 2: An alternative way to generalize H is

$$H'_{GC} \triangleq \left[\begin{array}{cccc|ccc} 1 & x^{i_2} & \cdots & x^{i_m} & x^{i_{m+1}} & \cdots & x^{i_{n_v}} \\ \hline & & M & & & I & \end{array}\right] \sim$$
$$\left[\begin{array}{ccc|ccc} 1 & \cdots & 1 & 1 & \cdots & 1 \\ \hline & M^{\uparrow\{1, x^{-i_2}, \ldots, x^{-i_m}\}} & & & I^{\uparrow\{x^{-i_{m+1}}, \ldots, x^{-i_{n_v}}\}} & \end{array}\right],$$

where $M^{\uparrow\{1, x^{-i_2}, \ldots, x^{-i_m}\}}$ and $I^{\uparrow\{x^{-i_{m+1}}, \ldots, x^{-i_{n_v}}\}}$ are the generalized (expanded) matrices of M and I respectively, lifted with the circulant order shown in the exponent, according to their sizes. Note that the two matrices H_{GC} and H'_{GC} are column-equivalent (equivalent after performing column operations, like dividing by a monomial) which means that they give two equivalent codes, so without any loss of generality, we work with H_{GC} in the sequel for simplicity. □

Example 4: Let $n_v = 6$, and

$$h_{GC} \triangleq \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad (9)$$

$$H_{GC} \triangleq \left[\begin{array}{ccc|ccc} 1 & x^{i_2} & x^{i_3} & x^{i_4} & x^{i_5} & x^{i_6} \\ \hline 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \quad (10)$$

$$\sim \left[\begin{array}{ccc|c} f_1 & f_2 & f_3 & \mathbf{0} \\ \hline & M & & I \end{array}\right], \quad (11)$$

where $f_1 \triangleq 1 + x^{i_4} + x^{i_5}$, $f_2 \triangleq x^{i_2} + x^{i_4} + x^{i_6}$, and $f_3 \triangleq x^{i_3} + x^{i_5} + x^{i_6}$. Let $g(x) \triangleq \gcd(f_1, f_2, f_3, x^N + 1)$ and $f(x)$ be such that $f(x)g(x) = x^N + 1$. Following Theorem 1, the matrix

$$G = \left[\begin{array}{c|c} G_{short} & G_{short}M^\mathsf{T} \end{array}\right] = \left[\begin{array}{ccc|ccc} f_3^\mathsf{T} & 0 & f_1^\mathsf{T} & f_3^\mathsf{T} & f_1^\mathsf{T}+f_3^\mathsf{T} & f_1^\mathsf{T} \\ 0 & f_3^\mathsf{T} & f_2^\mathsf{T} & f_3^\mathsf{T} & f_2^\mathsf{T} & f_2^\mathsf{T}+f_3^\mathsf{T} \\ f^\mathsf{T} & 0 & 0 & f^\mathsf{T} & f^\mathsf{T} & 0 \\ 0 & f^\mathsf{T} & 0 & f^\mathsf{T} & 0 & f^\mathsf{T} \\ 0 & 0 & f^\mathsf{T} & 0 & f^\mathsf{T} & f^\mathsf{T} \end{array}\right]$$

forms a generator matrix for $\ker(H_{GC})$.

For example, for $N = 79$, and exponents $[i_2, i_3, i_4, i_5, i_6] = [54, 66, 71, 55, 69]$ given in [5], for which H has girth 12,[4] we compute f_i and obtain $f_1 = 1 + x^{71} + x^{55}, f_2 = x^{54} + x^{71} + x^{69}, f_3 = x^{66} + x^{55} + x^{69}$. Since $\gcd(f_3, x^{79} + 1) = 1$ in $\mathbb{F}_2[x]$, the last three rows of the matrix G are omitted. The polynomial generator matrix G has each row of weight 16, which is the minimum distance of the code. Therefore, we obtain a $[474, 158, 16]$ code with a generator matrix of the minimum possible weight. □

Example 5: Let $n_v = 7$ and

$$h_{GC} \triangleq \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} M & I \end{bmatrix},$$

$$H_{GC} \sim \left[\begin{array}{cccc|c} f_1 & f_2 & f_3 & f_4 & \mathbf{0} \\ \hline & & M & & I \end{array}\right],$$

where $f_1 = 1 + x^{i_5} + x^{i_6} + x^{i_7}, f_2 = x^{i_2} + x^{i_5} + x^{i_6}, f_3 = x^{i_3} + x^{i_5} + x^{i_7}, f_4 = x^{i_4} + x^{i_6} + x^{i_7}$. Following Theorem 1, the matrix G_{GC} below forms the main part of a polynomial generator matrix for the generalized code $\ker(H_{GC})$, to which rows containing the polynomial $f(x)$ must added if the rank is not full,

$$G_{GC} = \left[\begin{array}{cccc|ccc} f_4^\mathsf{T} & 0 & 0 & f_1^\mathsf{T} & f_4^\mathsf{T} & f_1^\mathsf{T}+f_4^\mathsf{T} & f_1^\mathsf{T}+f_4^\mathsf{T} \\ 0 & f_4^\mathsf{T} & 0 & f_2^\mathsf{T} & f_4^\mathsf{T} & f_2^\mathsf{T}+f_4^\mathsf{T} & f_2^\mathsf{T} \\ 0 & 0 & f_4^\mathsf{T} & f_3^\mathsf{T} & f_4^\mathsf{T} & f_3^\mathsf{T} & f_3^\mathsf{T}+f_4^\mathsf{T} \end{array}\right].$$

The weight (16 in this case) of the rows of the matrix G_{GC} is an upper bound to the minimum distance of the GLDPC code.

For example, let $N = 68$ and set the exponents $[i_2, i_3, i_4, i_5, i_6, i_7] = [61, 49, 44, 1, 46, 14]$, such that H has girth 12 [5]. We compute f_i and obtain $f_1 = 1 + x + x^{46} + x^{14}, f_2 = x^{61} + x + x^{46}, f_3 = x^{49} + x + x^{14}, f_4 = x^{44} + x^{46} + x^{14}$. Since $\gcd(f_4, x^{68} + 1) = 1$ in $\mathbb{F}_2[x]$, f_4 is invertible and the matrix G is a polynomial generator matrix of the code. Therefore, we obtain a $[476, 204, 16]$ code with a generator matrix of the minimum possible weight.[5] We denote this code C_1 and will present simulation results for it later in Section V-D. □

B. Direct Generalization of Both Constraints

The following theorem considers the case when both constraints are generalized.

[4] How to choose exponents to achieve the desired girth in the constraint matrix is given in [30].

[5] This is demonstrated for a code with constraint degree $n_v = 15$ in [31, Appendix 1].

Corollary 2: Let H be a $2 \times n_v$ polynomial constraint matrix like in (5) and let $h_{GC} = \begin{bmatrix} M_{n\times m} & I_{n\times n} \end{bmatrix}$ and $h'_{GC} = \begin{bmatrix} M_1 & M_2 \end{bmatrix}$, where M, M_1, and M_2 are $n \times m$, $k \times m$, and $k \times n$ matrices, respectively, I is the $n \times n$ identity matrix, and $n_v = m+n$. Here, h'_{GC} and h_{GC} correspond to the two generalized constraints. Let H_{GC} be the overall parity-check matrix of the GLDPC code based on H, h'_{GC}, and h_{GC} as

$$H_{GC} \triangleq \left[\begin{array}{c|c} M_1^{\uparrow\{1 \quad x^{i_2} \quad \cdots \quad x^{i_m}\}} & M_2^{\uparrow\{x^{i_{m+1}} \quad \cdots \quad x^{i_{n_v}}\}} \\ \hline M & I \end{array}\right]. \tag{12}$$

Let $H_{short} \triangleq M_1^{\uparrow\{1 \quad x^{i_2} \quad \cdots \quad x^{i_m}\}} + M_2^{\uparrow\{x^{i_{m+1}} \quad \cdots \quad x^{i_{n_v}}\}} M$, a $k \times m$ matrix, and let G_{short} be a generator matrix for $\ker(H_{short})$. Then the matrix $G = \left[\begin{array}{c|c} G_{short} & G_{short}M^\mathsf{T} \end{array}\right]$ is a generator matrix for $\ker(H_{GC})$ (to which some extra vectors need to be included, if it is not full rank).

Proof: After performing row operations on (12) that use the identity I to make a zero matrix above it, we obtain the following row-equivalent matrix which is a parity-check matrix for the same code

$$H_{GC} \sim \left[\begin{array}{c|c} H_{short} & \mathbf{0} \\ \hline M & I \end{array}\right].$$

Following Theorem 3, the matrix G forms a generator matrix for $\ker(H_{GC})$. ■

Example 4 (cont.). Let $n_v = 6$, and

$$h_{GC} \triangleq \begin{bmatrix} M & I \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix},$$

$h'_{GC} = \begin{bmatrix} M_1 & M_2 \end{bmatrix}$, and

$$H_{GC} \triangleq \left[\begin{array}{c|c} M_1^{\uparrow\{1 \quad x^{i_2} \quad x^{i_3}\}} & M_2^{\uparrow\{x^{i_4} \quad x^{i_5} \quad x^{i_6}\}} \\ \hline M & I \end{array}\right] \sim \left[\begin{array}{c|c} H_{short} & \mathbf{0} \\ \hline M & I \end{array}\right],$$

where h'_{GC} is a column permutation of h_{GC},[6] and

$$H_{short} \triangleq M_1^{\uparrow\{1 \quad x^{i_2} \quad x^{i_3}\}} + M_2^{\uparrow\{x^{i_4} \quad x^{i_5} \quad x^{i_6}\}} M.$$

For example, for $h'_{GC} = \begin{bmatrix} I & M \end{bmatrix}$ and H_{GC} as in (13), as shown at the bottom of the next page,

$$H_{short} \triangleq I^{\uparrow\{1 \quad x^{i_2} \quad x^{i_3}\}} + M^{\uparrow\{x^{i_4} \quad x^{i_5} \quad x^{i_6}\}} M = \begin{bmatrix} 1 + x^{i_4} + x^{i_5} & x^{i_4} & x^{i_5} \\ x^{i_4} & x^{i_2} + x^{i_4} + x^{i_6} & x^{i_6} \\ x^{i_5} & x^{i_6} & x^{i_3} + x^{i_5} + x^{i_6} \end{bmatrix}.$$

The matrix H_{short} is not invertible, therefore, $\ker(H_{short}) \neq \emptyset$, and we use Lemma 2 to obtain the generator matrix of very low rank. We could instead disregard one of the first three rows, and form a better code. □

Example 5 (cont.). In this example, we demonstrate the effect of the choices of M_1 and M_2. Let $n_v = 7$ and

$$h_{GC} \triangleq \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} M & I \end{bmatrix},$$

$h'_{GC} = \begin{bmatrix} M_1 & M_2 \end{bmatrix}$, and

[6] Note that taking one constraint code as a column permutation of another is a well established technique to obtain good GLDPC codes, see, e.g., [7], [8].

$$H_{GC} \triangleq \left[\begin{array}{c|c} M_1^{\uparrow\{x^{i_1}\ \ x^{i_2}\ \ x^{i_3}\ \ x^{i_4}\}} & M_2^{\uparrow\{x^{i_5}\ \ x^{i_6}\ \ x^{i_7}\}} \\ \hline M & I \end{array}\right]$$

$$\sim \left[\begin{array}{c|c} H_{short} & \mathbf{0} \\ \hline M & I \end{array}\right], \text{where}$$

$$H_{short} \triangleq M_1^{\uparrow\{x^{i_1}\ \ x^{i_2}\ \ x^{i_3}\ \ x^{i_4}\}} + M_2^{\uparrow\{x^{i_5}\ \ x^{i_6}\ \ x^{i_7}\}} M.$$

The matrices M_1 and M_2 can be any taken in many different ways. First, consider the choices below and the corresponding H_{short}

$$M_1 \triangleq \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad M_2 \triangleq \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix},$$

$$H_{short} =$$

$$\begin{bmatrix} x^{i_1} + x^{i_5} + x^{i_6} & x^{i_5} + x^{i_6} & x^{i_5} & x^{i_4} + x^{i_6} \\ x^{i_5} + x^{i_7} & x^{i_2} + x^{i_5} & x^{i_5} + x^{i_7} & x^{i_4} + x^{i_7} \\ x^{i_6} + x^{i_7} & x^{i_6} & x^{i_3} + x^{i_7} & x^{i_4} + x^{i_6} + x^{i_7} \end{bmatrix}$$

$$\sim \begin{bmatrix} x^{i_1} & x^{i_2} & x^{i_3} & x^{i_4} \\ x^{i_5} + x^{i_7} & x^{i_2} + x^{i_5} & x^{i_5} + x^{i_7} & x^{i_4} + x^{i_7} \\ x^{i_1} + x^{i_5} + x^{i_6} & x^{i_5} + x^{i_6} & x^{i_5} & x^{i_4} + x^{i_6} \end{bmatrix}.$$

Following Theorem 1, the matrix $G_{GC} = \begin{bmatrix} G_{short} & G_{short} M^{\mathsf{T}} \end{bmatrix}$ forms the polynomial generator matrix for the generalized code $\ker(H_{GC})$, where

$$G_{short} = \begin{bmatrix} \Delta_{234}^{\mathsf{T}} & \Delta_{134}^{\mathsf{T}} & \Delta_{124}^{\mathsf{T}} & \Delta_{123}^{\mathsf{T}} \end{bmatrix},$$

to which extra rows need to be added if the matrix is not full rank.

For example, let $N = 68$ with exponents $[i_1, i_2, \ldots, i_7] = [0, 61, 49, 44, 1, 46, 14]$ chosen to give a 2×7 matrix with the associated Tanner graph having girth 12. We compute Δ_{ijk} and obtain a weight 52 codeword in $\ker(H_{short})$, which is the possible minimum distance of this code ($34 \leq d \leq 52$ from Magma computations). Explicitly,

$$\begin{aligned} \Delta_{234} &= x^{64} + x^{59} + x^{53} + x^{51} + x^{41} + x^{40} + x^{38} + x^{36} \\ &\quad + x^{28} + x^{23} + x^{20} + x^{18} + x^{8} + x^{3}, \\ \Delta_{134} &= x^{64} + x^{63} + x^{60} + x^{47} + x^{45} + x^{39} + x^{36} + x^{28} \\ &\quad + x^{25} + x^{23} + x^{15} + x^{3}, \\ \Delta_{124} &= x^{60} + x^{59} + x^{51} + x^{47} + x^{45} + x^{40} + x^{39} + x^{38} \\ &\quad + x^{37} + x^{36} + x^{22} + x^{15} + x^{8} + x^{7}, \\ \Delta_{123} &= x^{64} + x^{60} + x^{53} + x^{50} + x^{47} + x^{43} + x^{42} + x^{41} \\ &\quad + x^{40} + x^{20} + x^{15} + x^{7}. \end{aligned}$$

Since $\gcd(\Delta_{234}, \Delta_{134}, \Delta_{124}, \Delta_{123}, x^{68}+1) = x^4 + 1$, in $\mathbb{F}_2[x]$, we obtain that G_{short} only has rank $N - 4 = 64$. We let $f(x)$ be defined such that $f(x)(x^4+1) = x^{68}+1$ in $\mathbb{F}_2[x]$, and add to this matrix G_{short}, any two of the codewords

$$(0, g_1, g_2, g_3), (g_3, g_4, g_5, 0), (g_1, 0, g_6, g_4), (g_2, g_6, 0, g_5),$$

where

$$\begin{aligned} g_1 &= f(x)\Delta_{1,3|3,4} = f(x)(x^{i_3}(x^{i_4} + x^{i_6}) + x^{i_4+i_5}), \\ g_2 &= f(x)\Delta_{1,3|2,4} = f(x)(x^{i_2}(x^{i_4} + x^{i_6}) + x^{i_4}(x^{i_5} + x^{i_6})), \\ g_3 &= f(x)\Delta_{1,3|2,3} = f(x)(x^{i_2+i_5} + x^{i_3}(x^{i_5} + x^{i_6})), \\ g_4 &= f(x)\Delta_{1,3|1,3} = f(x)(x^{i_1+i_5} + x^{i_3}(x^{i_1} + x^{i_5} + x^{i_6})), \\ g_5 &= f(x)\Delta_{1,3|1,2} = f(x)(x^{i_1}(x^{i_5} + x^{i_6}) + x^{i_2}(x^{i_1} + x^{i_5} + x^{i_6})), \\ g_6 &= f(x)\Delta_{1,3|1,4} = f(x)(x^{i_1}(x^{i_4} + x^{i_6}) + x^{i_4}(x^{i_1} + x^{i_5} + x^{i_6})). \end{aligned}$$

Then the matrix G is a polynomial generator matrix of the code. Together with the following three components of total weight 36,

$$\begin{aligned} (G_{short}M^{\mathsf{T}})[1] &= x^{63} + x^{53} + x^{41} + x^{37} + x^{36} + x^{25} + x^{22} \\ &\quad + x^{20} + x^{18} + x^{7}, \\ (G_{short}M^{\mathsf{T}})[2] &= x^{64} + x^{63} + x^{59} + x^{51} + x^{50} + x^{45} + x^{43} \\ &\quad + x^{42} + x^{39} + x^{38} + x^{25} + x^{18} + x^{8} + x^{7}, \\ (G_{short}M^{\mathsf{T}})[3] &= x^{50} + x^{45} + x^{43} + x^{42} + x^{40} + x^{39} + x^{37} \\ &\quad + x^{28} + x^{23} + x^{22} + x^{18} + x^{3}, \end{aligned}$$

we obtain a codeword in $\ker H_{GC}$ of weight 88, which may be the minimum distance of the code (the Magma program has $64 \leq d \leq 88$). Therefore, we obtain a $[476, 72, 64 \leq d \leq$

$$H_{GC} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & x^{i_4} & x^{i_5} & 0 \\ 0 & x^{i_2} & 0 & x^{i_4} & 0 & x^{i_6} \\ 0 & 0 & x^{i_3} & 0 & x^{i_5} & x^{i_6} \\ \hline 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 + x^{i_4} + x^{i_5} & x^{i_4} & x^{i_5} & 0 & 0 & 0 \\ x^{i_4} & x^{i_2} + x^{i_4} + x^{i_6} & x^{i_6} & 0 & 0 & 0 \\ x^{i_5} & x^{i_6} & x^{i_3} + x^{i_5} + x^{i_6} & 0 & 0 & 0 \\ \hline 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]. \tag{13}$$

$$H_{short} \sim \left[\begin{array}{c|ccc} 1 & x^{i_2-i_1} & x^{i_3-i_1} & x^{i_4-i_1} \\ \hline 0 & x^{i_2} + x^{i_5} + (x^{i_5} + x^{i_7})x^{i_2-i_1} & x^{i_5} + x^{i_7} + (x^{i_5} + x^{i_7})x^{i_3-i_1} & x^{i_4} + x^{i_7} + (x^{i_5} + x^{i_7})x^{i_4-i_1} \\ 0 & x^{i_5} + x^{i_6} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_2-i_1} & x^{i_5} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_3-i_1} & x^{i_4} + x^{i_6} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_4-i_1} \end{array}\right]. \tag{14}$$

$$\begin{aligned} H_{short,2} &\triangleq \begin{bmatrix} x^{i_2} + x^{i_5} + (x^{i_5} + x^{i_7})x^{i_2-i_1} & x^{i_5} + x^{i_7} + (x^{i_5} + x^{i_7})x^{i_3-i_1} & x^{i_4} + x^{i_7} + (x^{i_5} + x^{i_7})x^{i_4-i_1} \\ x^{i_5} + x^{i_6} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_2-i_1} & x^{i_5} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_3-i_1} & x^{i_4} + x^{i_6} + (x^{i_1} + x^{i_5} + x^{i_6})x^{i_4-i_1} \end{bmatrix} \\ &= \begin{bmatrix} x^{62} + x^{61} + x^{7} + x & x^{63} + x^{50} + x^{14} + x & x^{58} + x^{45} + x^{44} + x^{14} \\ x^{62} + x^{61} + x^{46} + x^{39} + x & x^{50} + x^{49} + x^{27} + x & x^{46} + x^{45} + x^{22} \end{bmatrix}. \end{aligned} \tag{15}$$

88] code. We denote this code as C_2 and will later provide simulation results for it in Section V-D.

Note that we could have also continued to simplify H_{short} (14), as shown at the bottom of the previous page, and thus consider the $[204, 72, 22 \le d \le 34]$ code given by $\ker(H_{short,2})$ (15), as shown at the bottom of the previous page. The following is a generator matrix of weight 34 (which is the minimum distance of this code)

$$G_{short,2} \triangleq \left[\begin{array}{ccc} f_1 & f_2 & f_3 \\ \hline f_4 & 0 & f_5 \\ f_6 & f_5 & 0 \end{array}\right],$$

where $f(x)(x^4+1) = x^{68}+1$ in $\mathbb{F}_2[x]$ and

$$\begin{aligned}
f_1 &= \Delta_{23} = x^{64}+x^{63}+x^{60}+x^{47}+x^{45}+x^{39}+x^{36}+x^{28} \\
&\quad +x^{25}+x^{23}+x^{15}+x^3, \\
f_2 &= \Delta_{13} = x^{60}+x^{59}+x^{51}+x^{47}+x^{45}+x^{40}+x^{39}+x^{38} \\
&\quad +x^{37}+x^{36}+x^{22}+x^{15}+x^8+x^7, \\
f_3 &= \Delta_{12} = x^{64}+x^{60}+x^{53}+x^{50}+x^{47}+x^{43}+x^{42}+x^{41} \\
&\quad +x^{40}+x^{20}+x^{15}+x^7, \\
f_4 &= f(x)(x^{i_4}+x^{i_6}+(x^{i_1}+x^{i_5}+x^{i_6})x^{i_4-i_1}) \\
&= x^{65}+x^{61}+x^{57}+x^{53}+x^{49}+x^{45}+x^{41}+x^{37}+x^{33} \\
&\quad +x^{29}+x^{25}+x^{21}+x^{17}+x^{13}+x^9+x^5+x, \\
f_5 &= f(x)(x^{i_5}+x^{i_6}+(x^{i_1}+x^{i_5}+x^{i_6})x^{i_2-i_1}) \\
&= x^{67}+x^{63}+x^{59}+x^{55}+x^{51}+x^{47}+x^{43}+x^{39}+x^{35} \\
&\quad +x^{31}+x^{27}+x^{23}+x^{19}+x^{15}+x^{11}+x^7+x^3, \\
f_6 &= f(x)(x^{i_5}+(x^{i_1}+x^{i_5}+x^{i_6})x^{i_3-i_1}) \\
&= x^{67}+x^{66}+x^{63}+x^{62}+x^{59}+x^{58}+x^{55}+x^{54}+x^{51} \\
&\quad +x^{50}+x^{47}+x^{46}+x^{43}+x^{42}+x^{39}+x^{38}+x^{35} \\
&\quad +x^{34}+x^{31}+x^{30}+x^{27}+x^{26}+x^{23}+x^{22}+x^{19} \\
&\quad +x^{18}+x^{15}+x^{14}+x^{11}+x^{10}+x^7+x^6+x^3+x^2.
\end{aligned}$$

The code generated by the weight 38 matrix $\begin{bmatrix} f_1 & f_2 & f_3 \end{bmatrix}$ has dimension 64 and minimum distance 38, and each row $\begin{bmatrix} f_4 & 0 & f_6 \end{bmatrix}$ and $\begin{bmatrix} f_5 & f_6 & 0 \end{bmatrix}$ adds a rank of 4, giving a total rank of 72, while $\begin{bmatrix} f_4 & 0 & f_6 \end{bmatrix}$ has weight 34, which is the minimum distance of $\ker(H_{short,2})$.

Other matrices M_1, M_2

Above, we chose a certain matrix h'_{GC} with the matrices M_1 and M_2, but this choice was arbitrary. There are many other ways to choose either of the two, and each might give a non-equivalent QC-GLDPC code. For example, the three versions below

$$\left[\begin{array}{c|c} M_1^{\uparrow}\{x^{i_1} \quad x^{i_2} \quad x^{i_3} \quad x^{i_4}\} & M_2^{\uparrow}\{x^{i_5} \quad x^{i_6} \quad x^{i_7}\} \end{array}\right]$$
$$= \begin{bmatrix} 1 & x^{61} & x^{49} & 0 & x & 0 & 0 \\ 1 & x^{61} & 0 & x^{44} & 0 & x^{46} & 0 \\ 1 & 0 & x^{49} & x^{44} & 0 & 0 & x^{14} \end{bmatrix},$$
$$\begin{bmatrix} x & 0 & 0 & 1 & x^{61} & x^{49} & 0 \\ 0 & x^{46} & 0 & 1 & x^{61} & 0 & x^{44} \\ 0 & 0 & x^{14} & 1 & 0 & x^{49} & x^{44} \end{bmatrix}, \text{and}$$
$$\begin{bmatrix} 0 & x & 0 & 0 & 1 & x^{61} & x^{49} \\ x^{44} & 0 & x^{46} & 0 & 1 & x^{61} & 0 \\ x^{44} & 0 & 0 & x^{14} & 1 & 0 & x^{49} \end{bmatrix},$$

give $[476, 71, 68 \le d \le 80]$, $[476, 69, 73 \le d \le 86]$, and $[476, 72, 66 \le d \le 88]$ codes, respectively. □

A short example demonstrating the procedure for larger constraint code weight is contained in [31, Appendix 1].

C. Generalizing After Observing Prelifting

For the 2×6 protograph, it was shown in [5] that the iterative decoding threshold was the closest to capacity when 75% of the simple check nodes of the lifted $2N \times 6N$ scalar matrix were generalized with (9). But this will, in most cases, break the QC structure and make the encoding less efficient. In order to maintain the QC structure, and possibly create a stronger code (with better minimum distance), we will apply some of our previous techniques from [32] concerning double liftings (a "pre-lift" and a "final lift"). We recall Theorem 49 from [30].

Theorem 4 ([30], Theorem 49): Every QC-LDPC code with parity-check matrix H of length $n_v N = n_v N_1 N_2$ is equivalent to a pre-lifted QC-LDPC code of with pre-lift size (first lifting factor) N_1 and circulant size (second lifting factor) N_2.

Proof: We can transform each polynomial entry $g(x) = g_0(x^{N_1}) + x g_1(x^{N_1}) + \cdots + x^{N_1-1} g_{N_1-1}(x^{N_1})$ of H into an $N_1 \times N_1$ equivalent matrix,

$$\left[g(x)\right] = \begin{bmatrix} g_0(x) & x g_{N_1-1}(x) & x g_{N_1-2}(x) & \cdots & x g_1(x) \\ g_1(x) & g_0(x) & x g_{N_1-1}(x) & \cdots & x g_{N_1-2}(x) \\ \vdots & \vdots & \cdots & \vdots & \\ g_{N_1-1}(x) & g_{N_1-2}(x) & g_{N_1-3}(x) & \cdots & g_0(x) \end{bmatrix}.$$

In the scalar matrix H we can see this equivalence by performing a sequence of column and row permutations (reordering of the columns and rows). We abuse the notation and use the equality sign between the two equivalent representations (which result in equivalent graphs). ■

For example, if $N_1 = 2$ and $N = 2N_2$, then the entries $x^{2a} = (x^2)^a$ and $x^{2a+1} = x(x^2)^a$ give the following transformations

$$\left[x^{2a}\right] = \begin{bmatrix} x^a & 0 \\ 0 & x^a \end{bmatrix} \quad \text{and} \quad \left[x^{2a+1}\right] = \begin{bmatrix} 0 & x^{a+1} \\ x^a & 0 \end{bmatrix}.$$

Similarly, if $N = 3N_2$, the transformations are

$$\left[x^{3a}\right] = \begin{bmatrix} x^a & 0 & 0 \\ 0 & x^a & 0 \\ 0 & 0 & x^a \end{bmatrix}, \left[x^{3a+1}\right] = \begin{bmatrix} 0 & 0 & x^{a+1} \\ x^a & 0 & 0 \\ 0 & x^a & 0 \end{bmatrix},$$
$$\text{and } \left[x^{3a+2}\right] = \begin{bmatrix} 0 & x^{a+1} & 0 \\ 0 & 0 & x^{a+1} \\ x^a & 0 & 0 \end{bmatrix},$$

with component matrices of size $N_2 \times N_2$.

Corollary 3: Let $i_1 = 2j_1, \ldots, i_m = 2j_m$ be the even exponents and $i_{m+1} = 2k_1 + 1, \ldots, i_{n_v} = 2k_n + 1$ the odd exponents, where $m + n = n_v$. Then the matrix

$$H \triangleq \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & x^{i_2} & x^{i_3} & \cdots & x^{i_{n_v}} \end{bmatrix}$$
$$= \begin{bmatrix} 1 & \cdots & 1 & 1 & \cdots & 1 \\ x^{2j_1} & \cdots & x^{2j_m} & x^{2k_1+1} & \cdots & x^{2k_n+1} \end{bmatrix}$$

is row and column equivalent to the matrix

$$\left[\begin{array}{ccc|ccc|ccc|ccc} 1 & \cdots & 1 & 0 & \cdots & 0 & 1 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & \cdots & 1 & 0 & \cdots & 0 & 1 & \cdots & 1 \\ \hline x^{j_1} & \cdots & x^{j_m} & 0 & \cdots & 0 & 0 & \cdots & 0 & x^{k_1+1} & \cdots & x^{k_n+1} \\ 0 & \cdots & 0 & x^{j_1} & \cdots & x^{j_m} & x^{k_1} & \cdots & x^{k_n} & 0 & \cdots & 0 \end{array}\right].$$

Let $h_{GC} = \begin{bmatrix} M_1 & M_2 \end{bmatrix}$ be the matrix used to generalize H, with M_1 an $\ell \times m$ matrix and M_2 an $\ell \times n$ matrix, for some chosen $\ell > 1$, and $m + n = n_v$. By generalizing the first three constraints of this matrix using h_{GC} and the lifting factor $N/2$, we obtain the parity-check matrix H_{GC} for a GLDPC code given in (16), as shown at the bottom of the page. (Here we again use the notation $A^{\uparrow\{j_i\}}$ for the generalized (expanded) matrix based on a matrix A, with the exponents reflecting the exponents in the generalized matrices.) If $\ell = n$ and $M_2 = I_{n\times n}$, as we will consider in some of our cases, and if we denote $M \triangleq M_1$, for simplicity, then this matrix is further equivalent to that (17), as shown at the bottom of the page, and then equivalent to

$$H_{GC} \triangleq \left[\begin{array}{cc|cc} M & 0 & I & 0 \\ 0 & M & 0 & I \\ \hline M^{\uparrow\{x^{j_1} \ \cdots \ x^{j_m}\}} & I^{\uparrow\{x^{k_1+1} \ \cdots \ x^{k_n+1}\}} M & 0 & 0 \\ \begin{bmatrix} x^{k_1} & \cdots & x^{k_n} \end{bmatrix} M & \begin{bmatrix} x^{j_1} & \cdots & x^{j_m} \end{bmatrix} & 0 & 0 \end{array}\right]$$
$$\sim \left[\begin{array}{c|c} \operatorname{diag}(M, M) & I_{2n\times 2n} \\ \hline H_{1,short} & 0_{(n+1)\times n} \end{array}\right],$$

where

$$H_{1,short} \triangleq \begin{bmatrix} M^{\uparrow\{x^{j_1} \ \cdots \ x^{j_m}\}} & I^{\uparrow\{x^{k_1+1} \ \cdots \ x^{k_n+1}\}} M \\ \begin{bmatrix} x^{k_1} & \cdots & x^{k_n} \end{bmatrix} M & \begin{bmatrix} x^{j_1} & \cdots & x^{j_m} \end{bmatrix} \end{bmatrix} \sim$$
$$\begin{bmatrix} \left(I^{\uparrow\{x^{k_1+1} \ \cdots \ x^{k_n+1}\}}\right)^{-1} M^{\uparrow\{x^{j_1} \ \cdots \ x^{j_m}\}} & M \\ \begin{bmatrix} x^{k_1} & \cdots & x^{k_n} \end{bmatrix} M & \begin{bmatrix} x^{j_1} & \cdots & x^{j_m} \end{bmatrix} \end{bmatrix}.$$

Following Theorem 1, a generator matrix for G for $\ker(H_{GC})$ is

$$\begin{bmatrix} G_{1,short} & G_{1,short}\operatorname{diag}(M^{\mathsf{T}}, M^{\mathsf{T}}) \end{bmatrix},$$

where $G_{1,short}$ is a generator matrix for $\ker(H_{1,short})$.

We will now show how to do this for the example of $n_v = 6$. In particular, we will show how to simplify $H_{1,short}$ in the same manner that we simplified H_{GC}.[7]

[7] An example of how to do this for $n_v = 7$ is included as [31, Appendix 2]. In this case, we note that since n_v is odd, and not double $n_c = 3$, the process is slightly different.

Example 4 (cont.). Consider again $n_v = 6$ and $[i_1, i_2, i_3, i_4, i_5, i_6] = [1, x^{54}, x^{66}, x^{71}, x^{55}, x^{69}]$ from earlier. First, we increase the lifting exponent to $N = 90$, which is the first even exponent for which the girth of the 2×6 constraint matrix is 12. Then, we generalize the first constraints using $h_{GC} \triangleq \begin{bmatrix} M & I \end{bmatrix}$ from the examples before and obtain $H_{1,short}$ (below).

The code $\ker(H_{1,short})$ is a $[270, 91, 27]$ code, its minimum distance is a lower bound on the distance of the original code. We will further simplify this matrix. $H_{1,short}$ is equivalent to

$$\left[\begin{array}{ccc|ccc} x^{35}+x^{27} & x^{35}+x^{34} & x^{27}+x^{34} & 1 & x^{27} & x^{33} \\ \hline x^{-36} & x^{27-36} & 0 & 1 & 1 & 0 \\ x^{-28} & 0 & x^{33-28} & 1 & 0 & 1 \\ 0 & x^{27-35} & x^{33-35} & 0 & 1 & 1 \end{array}\right]$$

which can be row reduced into an equivalent matrix (18), as shown at the bottom of the next page, where

$$B^{-1} = (1 + x^{27} + x^{33})^{-1} \begin{bmatrix} 1 & x^{27} & x^{33} \\ 1 & 1 + x^{33} & x^{33} \\ 1 & x^{27} & 1 + x^{27} \end{bmatrix}$$

and from which $B^{-1}A$ can be computed as (19), as shown at the bottom of the next page. We do not need to substitute $(1+x^{27}+x^{33})^{-1} = x^{30}+x^{27}+x^{18}+x^{15}+x^{12}+x^{9}+x^{6}+x^{3}+1$. Following Theorem 1, if $G_{2,short}$ is a polynomial generator matrix for the code $\ker(H_{2,short})$, where

$$H_{2,short} \triangleq \begin{bmatrix} x^{-36} + x^{-28} & x^{27}(x^{-35}+x^{-36}) & x^{33}(x^{-35} + x^{-28}) \end{bmatrix},$$

then $G_{1,short} \triangleq \begin{bmatrix} G_{2,short} & G_{2,short}(B^{-1}A)^{\mathsf{T}} \end{bmatrix}$ is a generator matrix for $\ker(H_{1,short})$. Therefore, (20) and the (sparser) (21), shown at the bottom of the next page, are generator matrices for the large code.[8]

[8] A generator matrix in this form allows for a very simple encoding, starting from an encoder G_2 which can be easily generated, either with Theorem 2 or using the Magma program. The polynomial $(1+x^{12}+x^{18})$ does not need to be multiplied through because polynomial multiplication is easily implemented in circuitry. Decoding could also be devised based on the dependence of the vectors $\mathbf{v}_2, \mathbf{v}_3$, and $\mathbf{v}_4$ on $\mathbf{v}_1$.

$$\left[\begin{array}{c|c|c|c} M_1 & 0 & M_2 & 0 \\ 0 & M_1 & 0 & M_2 \\ {M_1}^{\uparrow\{x^{j_1} \ \cdots \ x^{j_m}\}} & 0 & 0 & {M_2}^{\uparrow\{x^{k_1+1} \ \cdots \ x^{k_n+1}\}} \\ 0 & x^{j_1} \ \cdots \ x^{j_m} & x^{k_1} \ \cdots \ x^{k_n} & 0 \end{array}\right]. \tag{16}$$

$$\left[\begin{array}{cc|cc} M & 0 & I & 0 \\ 0 & M & 0 & I \\ \hline M^{\uparrow\{x^{j_1} \ \cdots \ x^{j_m}\}} & 0 & 0 & I^{\uparrow\{x^{k_1+1} \ \cdots \ x^{k_n+1}\}} \\ 0 & x^{j_1} \ \cdots \ x^{j_m} & x^{k_1} \ \cdots \ x^{k_n} & 0 \end{array}\right]. \tag{17}$$

For example, using Theorem 2, we obtain the following polynomial generator matrix $G_{2,short}$ for the rate 91/135 code $\ker(H_{2,short})$,

$$G_{2,short} = \begin{bmatrix} x^{-27}(x^{35}+x^{36}) & x^{36}+x^{28} & 0 \\ x^{-33}(x^{35}+x^{28}) & 0 & x^{36}+x^{28} \\ f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & f \end{bmatrix},$$

where $f(x)$ is defined such that $f(x)(1+x) = 1+x^{45}$. The matrix $G_{2,short}$, however, does not contain as a row a codeword of the smallest weight 3. We can easily find such a codeword, due to the simple structure of the matrix $H_{2,short}$. We can visually check that $u(x)\begin{bmatrix}1 & x^{27} & x^{33}\end{bmatrix}$ is a codeword, for every polynomial $u(x)$. Then a codeword in $\ker(H_{GC})$ is given by (22), shown at the bottom of the next page, where $v(x) = (1+x^{27}+x^{33})^{-1}(x^{44}+x^{38}+x^{37}+x^{18}+x^{10}+x^{6})$.

Taking $u(x) = (1+x^{27}+x^{33})$, we obtain a codeword with $v(x) = x^{44}+x^{38}+x^{37}+x^{18}+x^{10}+x^{6}$ of weight 6, giving a total vector $\mathbf{v}$ of weight 39, equal to the minimum distance. The part $(\mathbf{v}_1,\mathbf{v}_2)$ is a codeword of smallest weight 27 in $\ker(H_{1,short})$, and the part $\mathbf{v}_1$ is a codeword in $\ker(H_{2,short})$ of weight 9. The resulting code is a $[540, 91, 39]$ code.

We can similarly obtain other vectors of small weight, starting from a different vector of weight 3, and thus obtain a polynomial matrix G of the smallest possible weight.[9] □

[9]Note that we can also find $\mathbf{v}$ directly from the matrix G_2 above. Since $\gcd(x^{28}+x^{36}, f(x)) = 1$ in $\mathbb{F}_2[x]$, there exists $u(x)$ and $v(x)$ such that $u(x)(x^{28}+x^{36})+v(x)f(x) = 1$, in $\mathbb{F}_2[x]$, where $u(x) = x^{42}+x^{39}+x^{38}+x^{36}+x^{35}+x^{33}+x^{32}+x^{29}+x^{26}+x^{23}+x^{22}+x^{20}+x^{19}+x^{17}+x^{16}+x^{13}+x^{10}+x^{9}+x^{7}+x^{6}+x^{4}+x^{3}+1$. Therefore, in $\mathbb{F}_2[x]/(x^{45}+1)$, we have that $u(x)(x^{28}+x^{36})+f(x) = 1$. If we take $\mathbf{u} \triangleq \left(x^{27}u(x), x^{33}u(x), 1, 1, 1\right)$, then $\mathbf{v} \triangleq \mathbf{u}G$ is the codeword of weight 39 that we obtain directly in the text above, with $u(x) = (1+x^{27}+x^{33})$.

We can also use different matrices for the generalized constraint nodes.

Example 4 (cont.). For example, we generalized the first two nodes using $h_{GC,0} \triangleq \begin{bmatrix} M & I \end{bmatrix}$ and the third using $h_{GC,1} \triangleq \begin{bmatrix} I & M \end{bmatrix}$ to obtain a matrix H_{GC}. Applying the same technique, and Theorem 1, we obtain that a generator matrix for the generalized code $\ker(H_{GC})$ is

$$\begin{bmatrix} G_{1,short} & G_{1,short}\operatorname{diag}(M^{\mathsf{T}}, M^{\mathsf{T}}) \end{bmatrix},$$

where $G_{1,short}$ is a generator matrix for the shorter code with parity-check matrix

$$H_{1,short} \triangleq \left[\begin{array}{c|c} I & \left(I^{\uparrow\{1,27,33\}}\right)^{-1} M^{\uparrow\{36,28,35\}} M \\ \begin{bmatrix} x^{35} & x^{27} & x^{34} \end{bmatrix} M & \begin{bmatrix} 1 & x^{27} & x^{33} \end{bmatrix} \end{array}\right] \sim \left[\begin{array}{c|ccc} I & & A & \\ \hline 0 & f_1 & f_2 & f_3 \end{array}\right],$$

where

$$\begin{aligned} f_1 &= 1+x^5+x^6+x^{17}+x^{22}+x^{29}+x^{33}, \\ f_2 &= x^4+x^6+x^{25}+x^{27}+x^{29}+x^{33}+x^{36}, \\ f_3 &= x^4+x^5+x^{17}+x^{22}+x^{25}+x^{33}+x^{36}, \text{ and} \\ A &\triangleq \begin{bmatrix} x^{36}+x^{28} & x^{36} & x^{28} \\ x^9 & x^9+x^8 & x^8 \\ x^{33} & x^2 & x^{33}+x^2 \end{bmatrix}. \end{aligned}$$

The code given by $H_{1,short}$ is a $[270, 90, 30]$ code. Applying Theorem 1, if $G_{2,short}$ is a polynomial generator matrix for the code $\ker(H_{2,short})$, where $H_{2,short} = \begin{bmatrix} f_1 & f_2 & f_3 \end{bmatrix}$, then $\begin{bmatrix} G_{2,short}A^{\mathsf{T}} & G_{2,short} \end{bmatrix}$ is a polynomial generator matrix for the code $\ker(H_{1,short})$, and thus,

$$\begin{bmatrix} G_{2,short}A^{\mathsf{T}} & G_{2,short} & G_{2,short}A^{\mathsf{T}}M^{\mathsf{T}} & G_{2,short}M^{\mathsf{T}} \end{bmatrix}$$

is a generator matrix for the original code, $\ker(H_{GC})$.

$$\begin{aligned} &\left[\begin{array}{ccc|ccc} x^{35}+x^{27} & x^{35}+x^{34} & x^{27}+x^{34} & 1 & x^{27} & x^{33} \\ x^{-36} & x^{27-36} & 0 & 1 & 1 & 0 \\ x^{-28} & 0 & x^{33-28} & 1 & 0 & 1 \\ \hline x^{-36}+x^{-28} & x^{27}(x^{-35}+x^{-36}) & x^{33}(x^{-35}+x^{-28}) & 0 & 0 & 0 \end{array}\right] = \\ &\left[\begin{array}{ccc|c} & A & & B \\ \hline x^{-36}+x^{-28} & x^{27}(x^{-35}+x^{-36}) & x^{33}(x^{-35}+x^{-28}) & 0 \end{array}\right] \\ &\sim \left[\begin{array}{ccc|c} & B^{-1}A & & I \\ \hline x^{-36}+x^{-28} & x^{27}(x^{-35}+x^{-36}) & x^{33}(x^{-35}+x^{-28}) & 0 \end{array}\right], \end{aligned} \tag{18}$$

$$B^{-1}A = (1+x^{27}+x^{33})^{-1} \begin{bmatrix} x^{36}+x^{35}+x^{27}+x^5 & x^{35}+x^{34}+x^{18} & x^{38}+x^{34}+x^{27} \\ x^{42}+x^{35}+x^{27}+x^9+x^5 & x^{36}+x^{35}+x^{34}+x^{24} & x^{38}+x^{34}+x^{27} \\ x^{44}+x^{36}+x^{35}+x^{27}+x^{17} & x^{35}+x^{34}+x^{18} & x^{34}+x^{32}+x^{27}+x^5 \end{bmatrix}. \tag{19}$$

$$G \triangleq \begin{bmatrix} G_{1,short} & G_{1,short}\operatorname{diag}(M^{\mathsf{T}}, M^{\mathsf{T}}) \end{bmatrix} = \begin{bmatrix} G_{2,short} & G_{2,short}(B^{-1}A)^{\mathsf{T}} & G_{2,short}M & G_{2,short}(B^{-1}A)^{\mathsf{T}}M^{\mathsf{T}} \end{bmatrix}. \tag{20}$$

$$G' \triangleq (1+x^{12}+x^{18})\begin{bmatrix} G_{2,short} & G_{2,short}(B^{-1}A)^{\mathsf{T}} & G_{2,short}M^{\mathsf{T}} & G_{2,short}(B^{-1}A)^{\mathsf{T}}M^{\mathsf{T}} \end{bmatrix}. \tag{21}$$

For example, using Theorem 2, we obtain the following polynomial generator matrix $G_{2,short}$ for the code $\ker H_{2,short}$ of minimum distance 9 as

$$G_2 = \begin{bmatrix} f_2^{\mathsf{T}} & f_1^{\mathsf{T}} & 0 \\ f_3^{\mathsf{T}} & 0 & f_1^{\mathsf{T}} \end{bmatrix}.$$

The rows of G_2 can be used to find codewords in the original code, but they do not lead to codewords of the smallest weight 40. A codeword of weight 40 can be obtained by using the structure of the codewords as follows. (We obtained them by reverse engineering.) Since $f_1 + f_2 + f_3 = 1 + x^{27} + x^{33}$ is invertible, we can take $\mathbf{v}_2 = (v, v, v + x^a)$ (because, when we multiply them with M, they will have as close to 0 weight as possible), such that

$$\begin{aligned} H_{2,short}\mathbf{v}_2^{\mathsf{T}} &= f_1 v^{\mathsf{T}}(x) + f_2 v^{\mathsf{T}}(x) + f_3 v^{\mathsf{T}}(x) + f_3 x^{-a} \\ &= (f_1 + f_2 + f_3)v^{\mathsf{T}}(x) + x^{-a} f_3 = 0. \end{aligned}$$

Taking $v^{\mathsf{T}}(x) = x^{-a}(f_1 + f_2 + f_3)^{-1} f_3$, we obtain a codeword for any a. We can also choose $\mathbf{v}_2 = (v + x^a, v, v)$ or $\mathbf{v}_2 = (v, v + x^a, v)$, and choose i such that $(f_1 + f_2 + f_3)^{-1} f_i$ has the smallest weight, for $i = 1, 2, 3$. In this case, it is for $i = 3$, so we take

$$\begin{aligned} &v^{\mathsf{T}}(x) = x^{-a}(f_1 + f_2 + f_3)^{-1} f_3 = \\ &x^{-a}(x^{36} + x^{32} + x^{22} + x^{21} + x^{16} + x^{10} + x^5 + x^4 + 1), \end{aligned}$$

because

$$\begin{aligned} &(f_1 + f_2 + f_3)^{-1} \\ &= x^{36} + x^{34} + x^{33} + x^{31} + x^{27} + x^{26} + x^{25} + x^{23} + x^{21} \\ &+ x^{18} + x^{15} + x^{14} + x^{13} + x^{12} + x^9 + x^8 + x^2 + x + 1. \end{aligned}$$

For example, $a = 1$ gives $\mathbf{v}_2 = (v, v, v + 1)$, with

$$v^{\mathsf{T}} = x^{36} + x^{32} + x^{22} + x^{21} + x^{16} + x^{10} + x^5 + x^4 + 1.$$

Then, the weight of $\mathbf{v}_4 = \mathbf{v}_2 M^{\mathsf{T}}$ is 2, for any a, and any position of x^{-a}, having a component equal to 0 and two equal to x^{-a}. Then $\mathbf{v}_1 = \mathbf{v}_2 A^{\mathsf{T}}$ is x^a times one of columns of A, depending on the component in which x^{-a} is placed in $\mathbf{v}_2^{\mathsf{T}}$, so it has weight 4, for any a. Finally, the weight of $\mathbf{v}_3 = \mathbf{v}_1 M^{\mathsf{T}}$ has maximum weight $\mathrm{wt}(\mathbf{v}_3) = 3 + 2 + 3 = 8$, giving a total of $3\mathrm{wt}(v) - 1 + 2 + 8 + 4 = 13 + 3\mathrm{wt}(v)$. For $a = 20$ and v above, we get $13 + 27 = 40$. □

An additional example of this procedure, continuing Example 5 is given in [31, Appendix 2].

D. Numerical Results

Fig. 1 presents some simulation results of two QC-GLDPC codes from Example 5, one partially generalized and one fully generalized. Specifically, code C_1 is a $[476, 204, 16]$ partially generalized QC-GLDPC code and code C_2 is a $[476, 72, 64 \le d \le 88]$ fully generalized QC-GLDPC code. Simulations were performed using belief propagation (BP) decoding on the non-generalized parity-check matrix with a maximum number of 100 iterations, where the SPC constraints are updated as the conventional sum-product algorithm (SPA), but the soft-outputs from the generalized constraints are obtained using the Bahl-Cocke-Jelinek-Raviv (BCJR) algorithm (see [7] for details). Log-likelihood ratios are clipped at ± 20 and for the BCJR algorithm we set the forward, backward, and branch metrics to have a threshold of $\pm 2.5 \times 10^4$. Also shown in the figure are the performance of the expanded parity-check matrix, corresponding to the same codes C_1 and C_2 but where we expand all the generalized constraints to SPCs and decode with conventional BP decoding for a maximum of 100 iterations. Finally, we also show the BP decoding performance of randomly-constructed $[476, 207, 16 \le d \le 62]$, girth 6, $(4,7)$-regular and $[476, 74, 62 \le d \le 102]$, girth 6, $(6,7)$-regular QC-LDPC codes that have similar code dimensions to our examples. Their respective parity-check matrices (lifted with $N = 68$) are given by

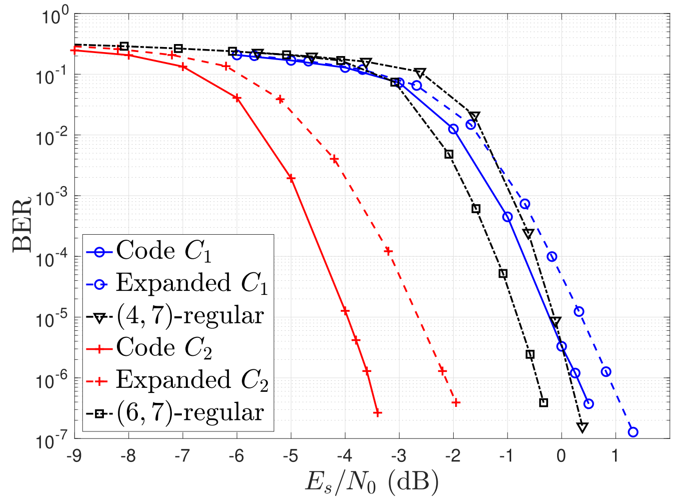

Fig. 1. Computer simulation results obtained for the partially generalized $[476, 204, 16]$ QC-GLDPC code C_1 and fully generalized $[476, 72, 64 \le d \le 88]$ QC-GLDPC code C_2 from Example 5.

$$\begin{bmatrix} x^7 & x^{48} & x^{56} & x^{10} & x^{61} & x^{16} & x^{52} \\ x^{65} & x^{34} & x^{17} & x^{40} & x^{37} & x^{30} & x^{59} \\ x^{29} & x^{22} & x^{36} & x^{63} & x^{29} & x^{36} & x^{49} \\ x^5 & x^{51} & x^{29} & x^7 & x^{10} & x^{24} & x^{27} \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & x^{23} & x^{47} & x^{27} & x^6 & x^{16} & x^{63} \\ 1 & x^{22} & x^{55} & x^{53} & x^{43} & x^{29} & x^6 \\ 1 & x^{16} & x^{39} & x^{51} & x^{46} & x^{17} & x^{27} \\ 1 & x^{12} & x^{21} & x^8 & x^{61} & x^{60} & x^{43} \\ 1 & x^{54} & x & x^2 & x^{15} & x^{22} & x^{53} \end{bmatrix}.$$

$$\begin{aligned} \mathbf{v} &= u(x)\begin{bmatrix} 1 & x^{27} & x^{33} \end{bmatrix}\begin{bmatrix} I & (B^{-1}A)^{\mathsf{T}} & M^{\mathsf{T}} & (MB^{-1}A)^{\mathsf{T}} \end{bmatrix} \\ &= u(x)\begin{bmatrix} 1 & x^{27} & x^{33} & 1 + x^{27} & 1 + x^{33} & x^{27} + x^{33} & v(x) & v(x) & v(x) & 0 & 0 & 0 \end{bmatrix}. \end{aligned} \quad (22)$$

As expected from the high girth and good minimum distance, the codes display a relatively steep waterfall and no indication of an error floor down to a bit error rate (BER) of 10^{-6}. For both C_1 and C_2, hybrid BP and BCJR decoding is seen to outperform BP decoding of the expanded parity-check matrices (both with girth 4), as expected. In the case of the fully generalized code C_2, the decoding performance of the QC-GLDPC code is significantly better than that of the random $(6, 7)$-regular QC-LDPC code with similar dimension. This is likely attributed to the large minimum distance of the generalized code as well as BP decoding advantages concerning the density and cycle distribution of the QC-LDPC code (the QC-GLDPC takes advantage of hybrid decoding on a girth 12 graph with BCJR component decoding, while the expanded graph has a sparse representation, although it has girth 4). However, although the partially generalized code has better performance than the random $(4, 7)$-regular QC-LDPC code at lower signal-to-noise ratios (SNRs), the random code has a steeper waterfall and the curves cross at around 1 dB. One possible reason for this is that there may not be enough generalization to sufficiently improve the minimum distance of this code with a $(2, 7)$-regular base graph (only 50% of constraints are generalized, less than the suggested 75% generalization from [5]).

VI. Concluding Remarks

This paper shows how to obtain a generator matrix for a QC-GLDPC code using minors of the polynomial parity-check matrix. The approach can be applied to fully generalized matrices, or partially generalized (with mixed constraint nodes) in order to find better performance/rate trade-offs. The resulting matrices can be presented in several forms that may facilitate efficient encoder implementation as well as minimum distance analysis. These forms allow us to display a variety of codewords and possibly, those of weight equal to the minimum distance. We also demonstrated that the code parameters can be improved by applying a double graph-lifting procedure that does not affect the ability to obtain a polynomial generator matrix.

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Roxana Smarandache (Senior Member, IEEE) received the B.Sc. degree in mathematics from the University of Bucharest, in 1996, and the M.Sc. and Ph.D. degrees from the University of Notre Dame, Notre Dame, IN, USA, in 1997 and 2001, respectively. After 11 years on the faculty of San Diego State University, she joined the University of Notre Dame, in 2012, where she has a joint appointment in mathematics and electrical engineering. From 2021 to 2022, she was on a Sabbatical Leave at the Swiss Federal Institute of Technology Zurich (ETHZ), from 2014 to 2015, the Swiss Federal Institute of Technology in Lausanne (EPFL), and the University of Zurich, Switzerland, from 2007 to 2008. She is currently a Professor of mathematics and electrical engineering with the University of Notre Dame. Her research interests include coding theory, combinatorics, and graph theory. In particular, she focuses on low density parity check codes, iterative and linear programming decoding, and convolutional codes. From 2022 to 2023, she was a Guest Editor of *Journal of Algebra and Its Applications* (JAA). From 2014 to 2017, she was an Associate Editor of IEEE TRANSACTIONS ON INFORMATION THEORY. From 2013 to 2014, she served as an Associate Editor for the *Advances in Mathematics of Communications* (AMC).

David G. M. Mitchell (Senior Member, IEEE) received the Ph.D. degree in electrical engineering from The University of Edinburgh, U.K., in 2009. From 2009 to 2015, he held post-doctoral research associate and visiting assistant professor positions with the Department of Electrical Engineering, University of Notre Dame, USA. Since 2015, he has been with the Klipsch School of Electrical and Computer Engineering, New Mexico State University, USA, where he is currently an Associate Professor and holds the IFT Professorship in Telecommunications. His research interests include digital communications, machine learning, and quantum computing, with emphasis on error control coding and information theory. He was a recipient of the National Science Foundation Faculty Early Career Award in 2022 and has received several best paper awards. He served as an Associate Editor for IEEE TRANSACTIONS ON INFORMATION THEORY from 2020 to 2024 and currently serves as an Associate Editor for IEEE TRANSACTIONS ON COMMUNICATIONS.

Anthony Gómez-Fonseca (Member, IEEE) received the B.Sc. degree in mathematics from the University of Puerto Rico at Cayey, in 2016, the M.Sc. degree in mathematics from the University of Puerto Rico Río Piedras Campus, in 2019, and the Ph.D. degree in mathematics from the University of Notre Dame in 2025. He is currently a Post-Doctoral Scholar with the University of South Florida. His research interests include coding theory, graph theory, and quantum error correction. In particular, he focuses on low-density parity-check codes.